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KIM et al.(10) **Pub. No.: US 2025/0287570 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **INTEGRATED CIRCUIT DEVICE**(71) Applicant: **Samsung Electronics Co., Ltd.**,
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Suwon-si (KR)(21) Appl. No.: **19/218,700**(22) Filed: **May 27, 2025****Related U.S. Application Data**(63) Continuation of application No. 17/734,436, filed on
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H10B 12/00 (2023.01)(52) **U.S. Cl.**
CPC **H10B 12/34** (2023.02); **H10B 12/053**
(2023.02)(57) **ABSTRACT**

An integrated circuit device includes a substrate including an active region defined by a device isolation layer, the substrate defining a gate trench extending across the active region, a gate dielectric layer conformally covering an inner surface of the gate trench, and a gate electrode filling the gate trench on the gate dielectric layer. The gate electrode is composed of crystal grains of a single metal, and a diagonal length of at least one of the crystal grains is greater than a height of the active region that is in contact with the gate electrode.

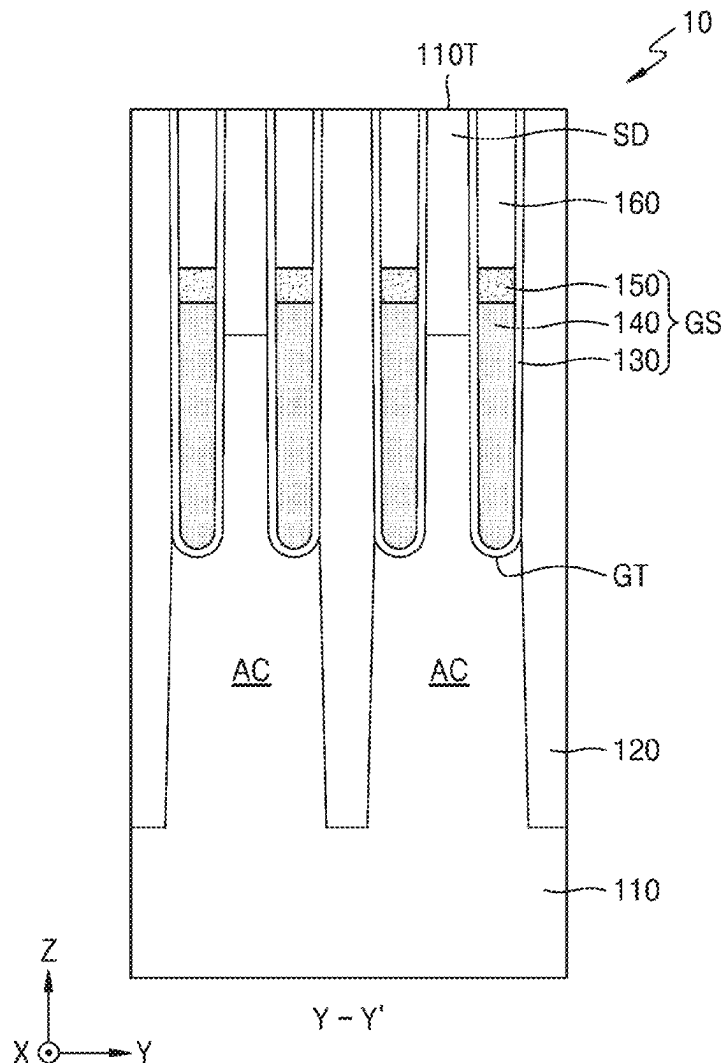


FIG. 1A

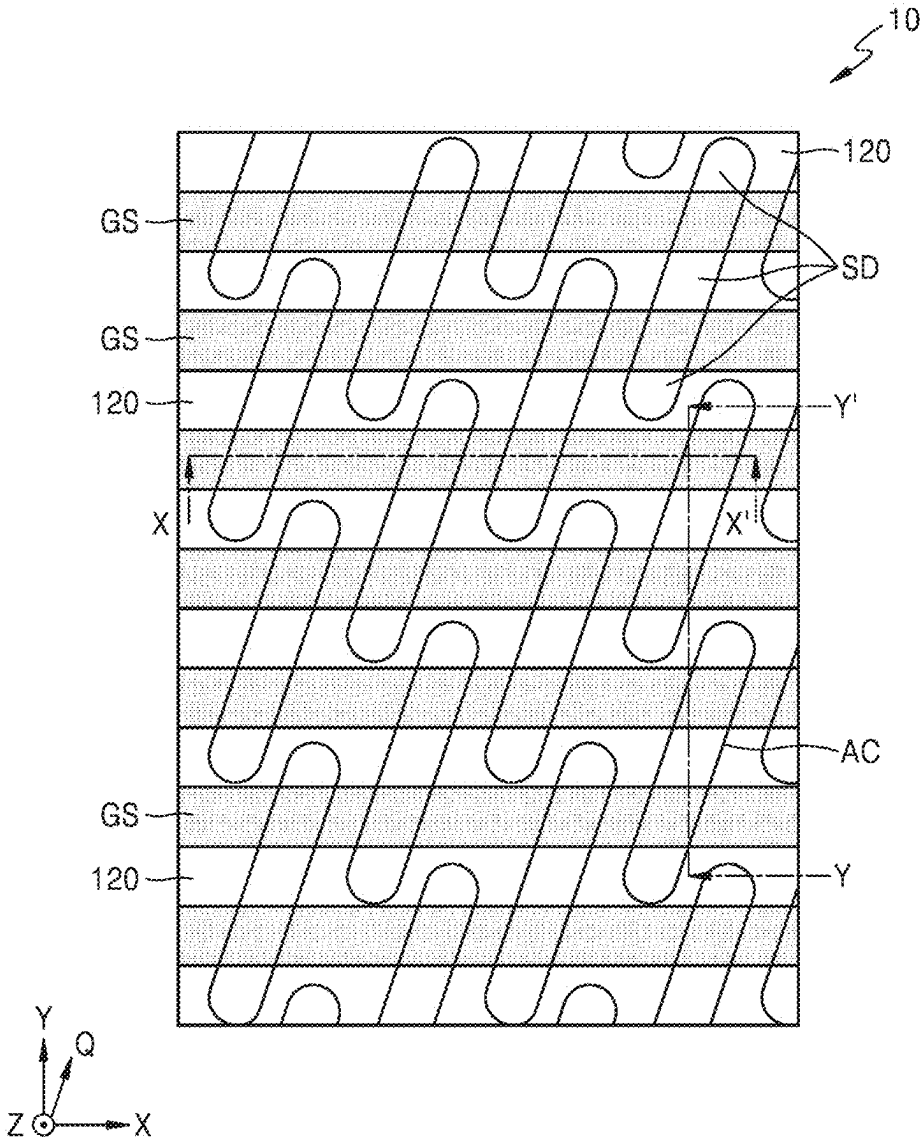


FIG. 1B

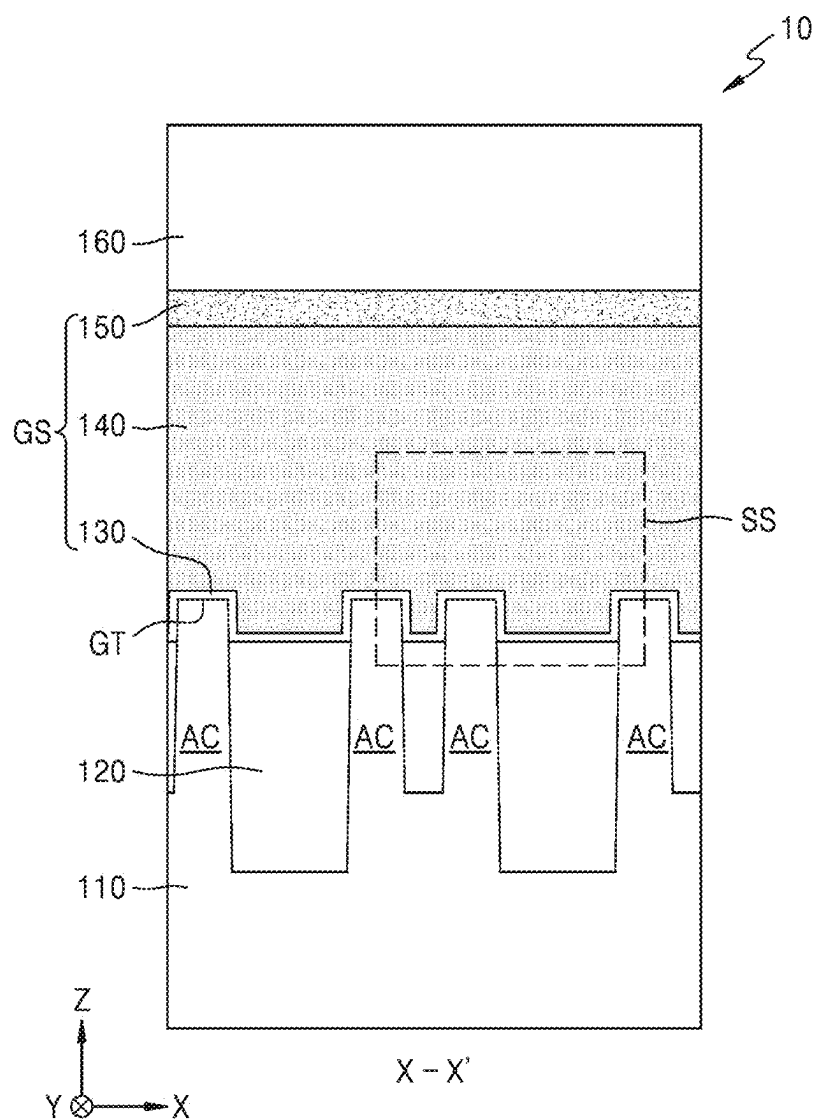


FIG. 1C

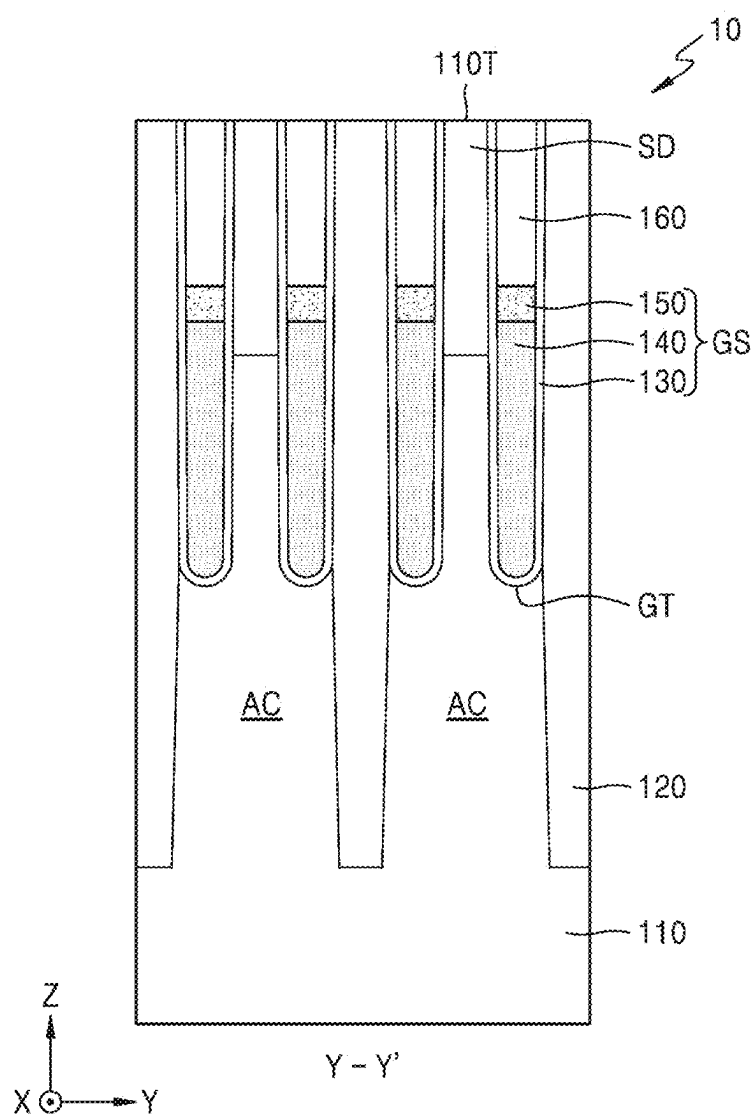


FIG. 1D

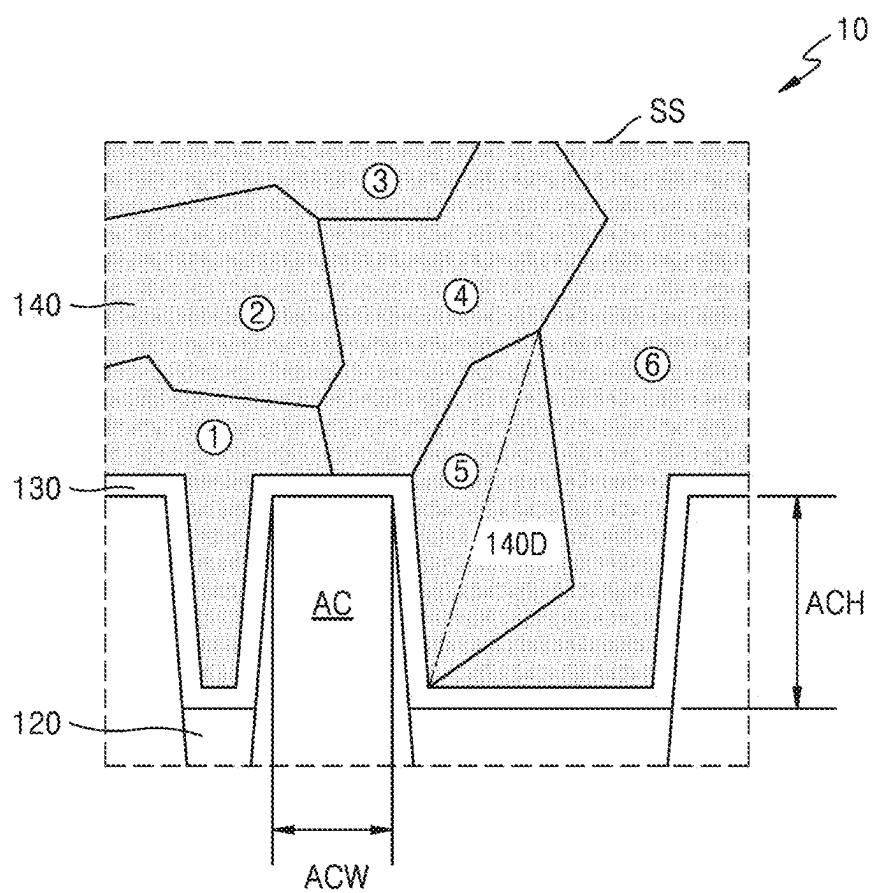


FIG. 2

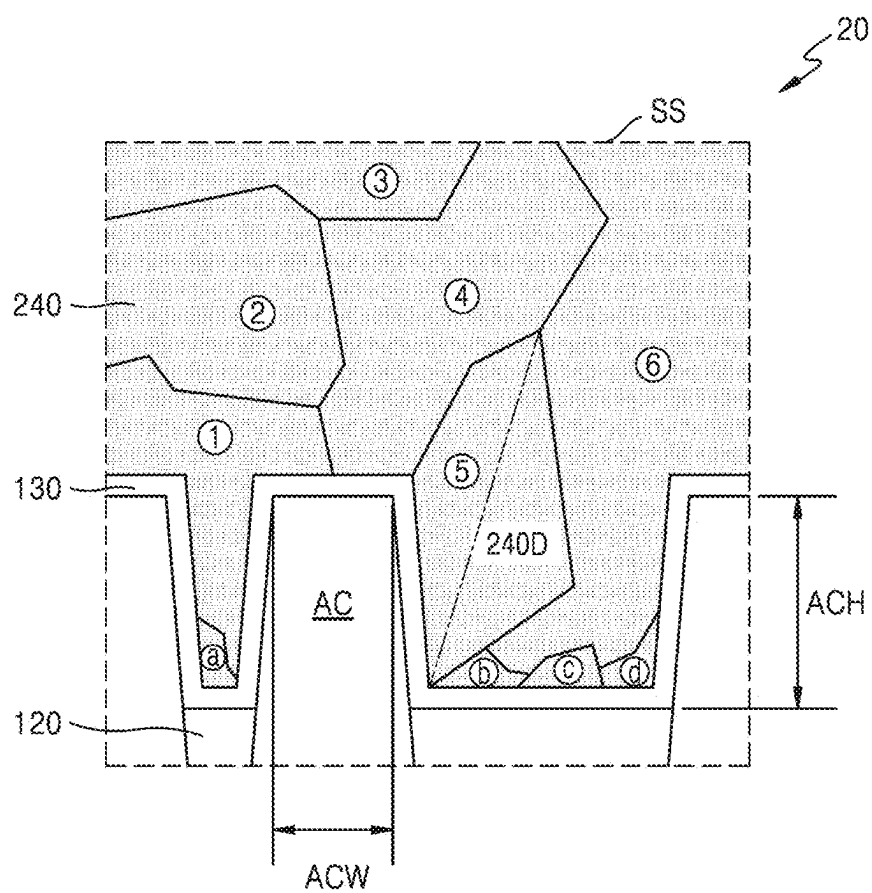


FIG. 3A

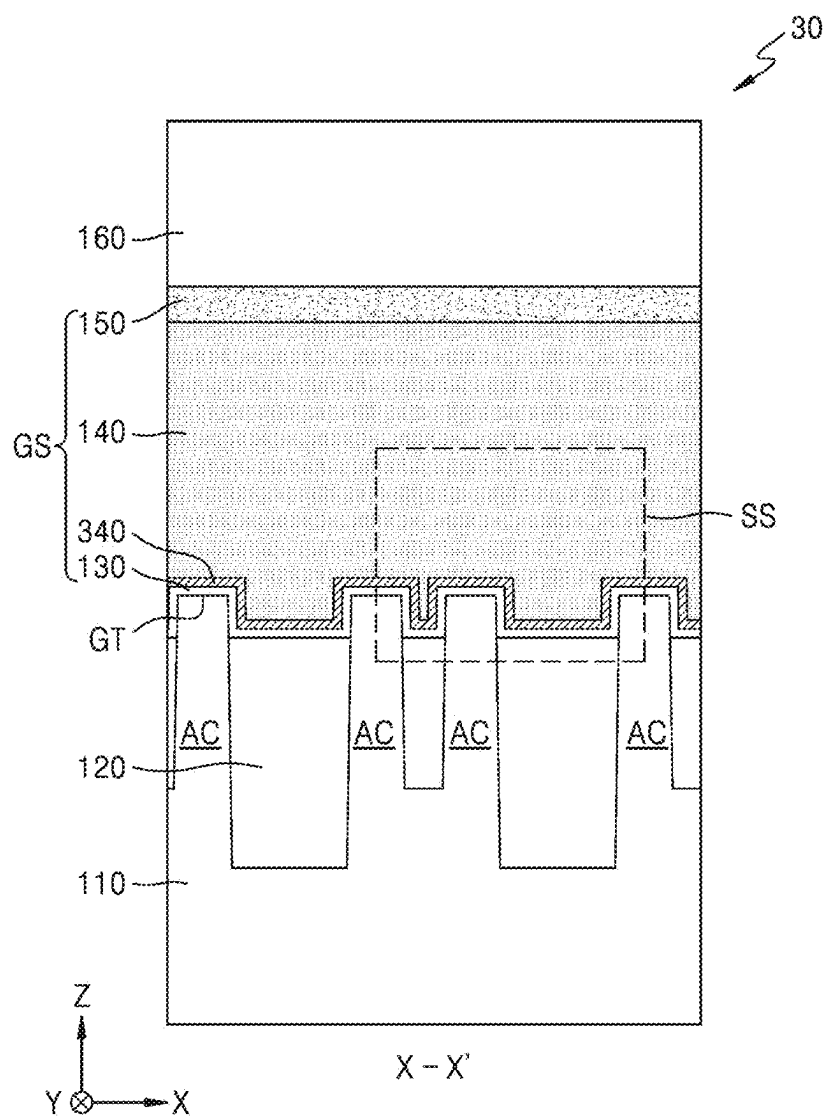


FIG. 3B

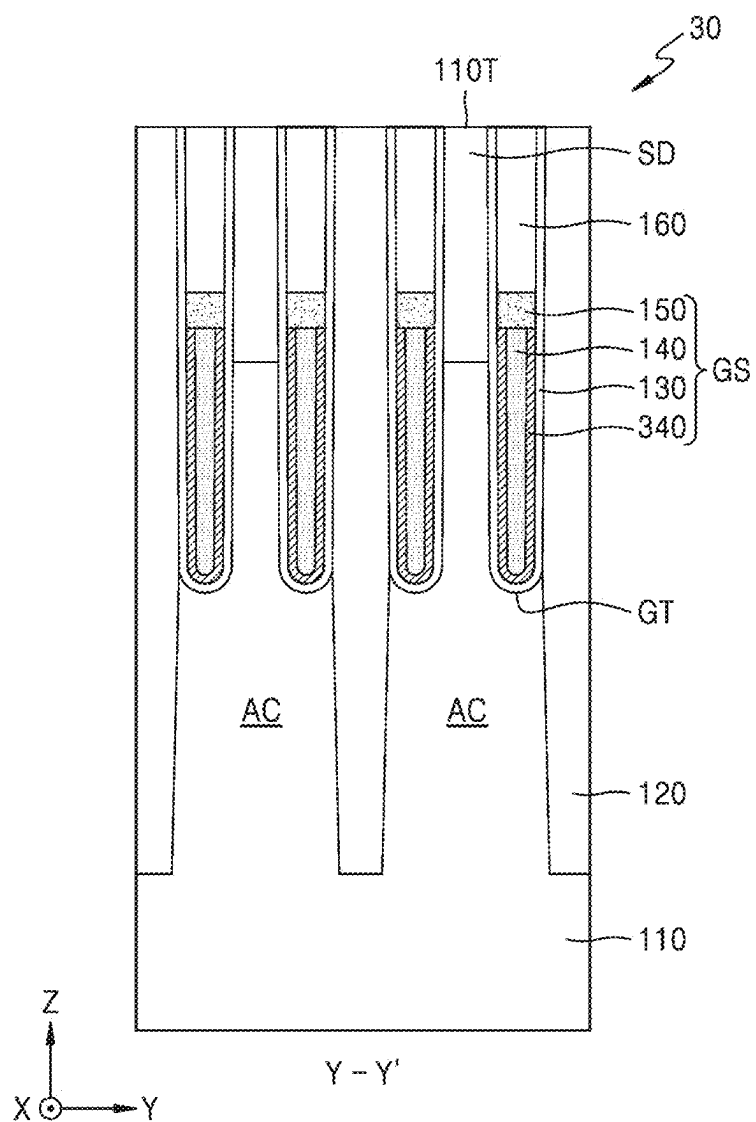


FIG. 3C

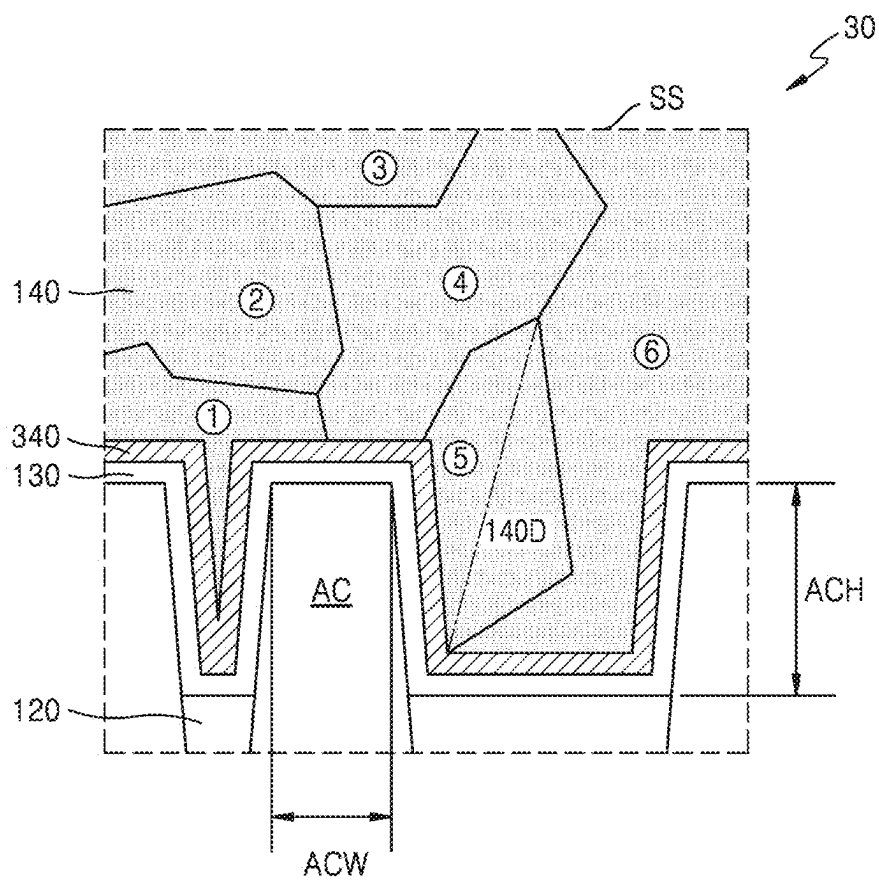


FIG. 4

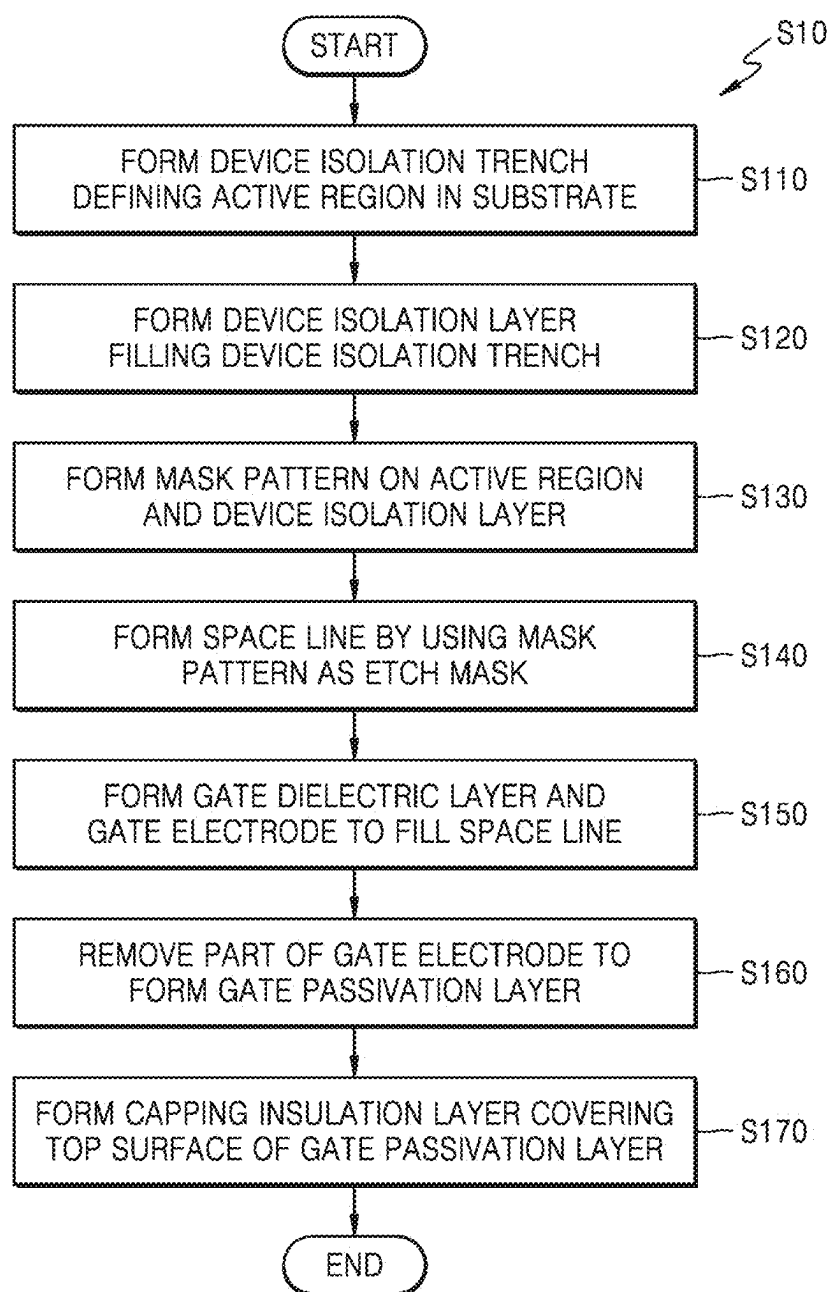


FIG. 5A

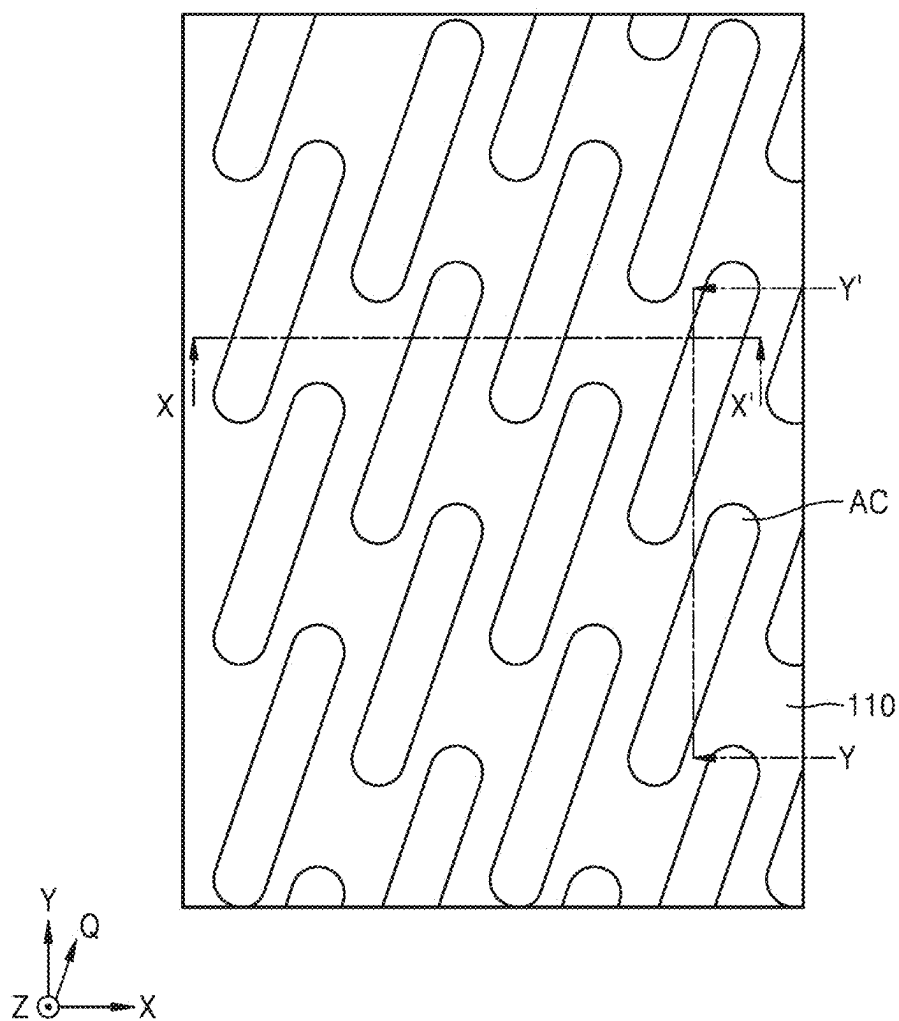


FIG. 5B

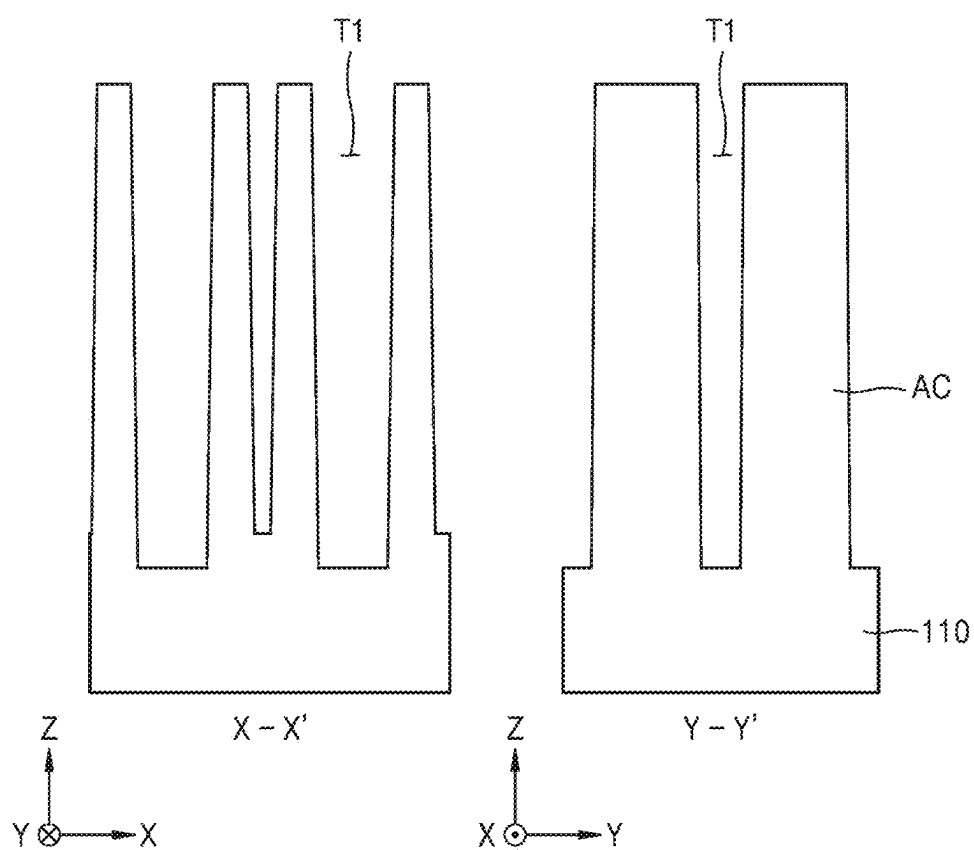


FIG. 6A

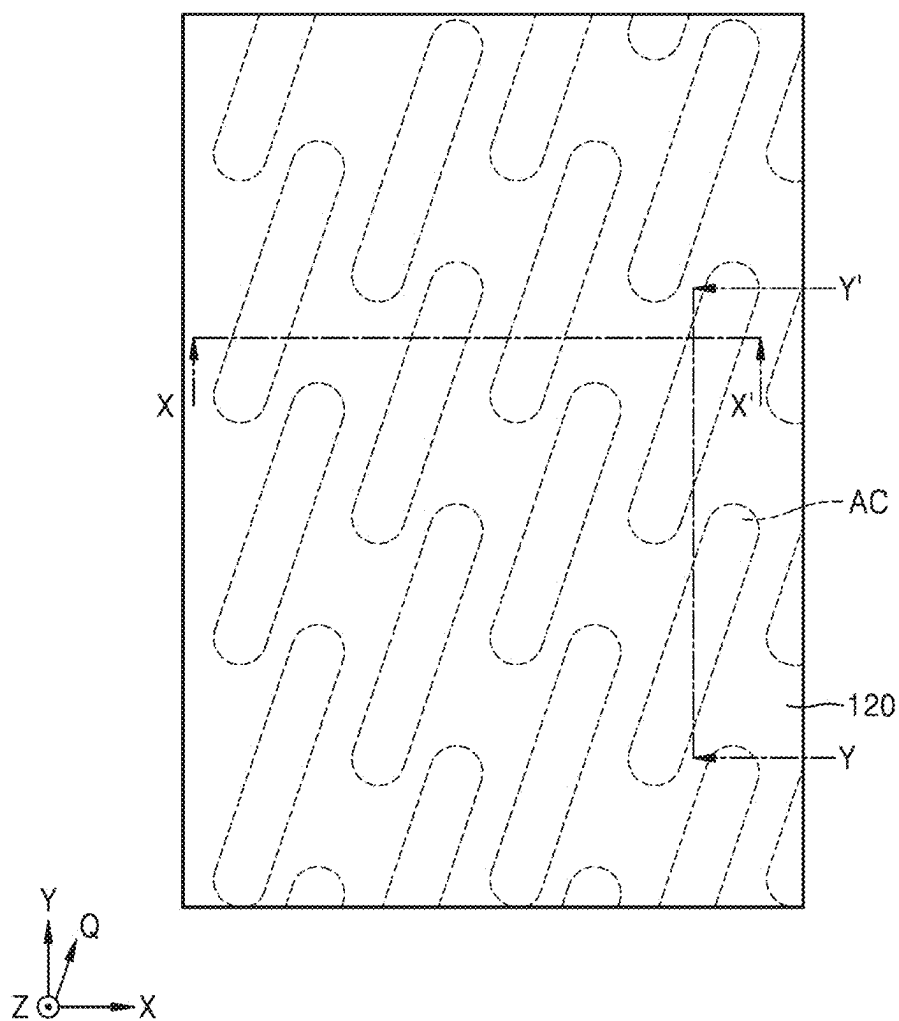


FIG. 6B

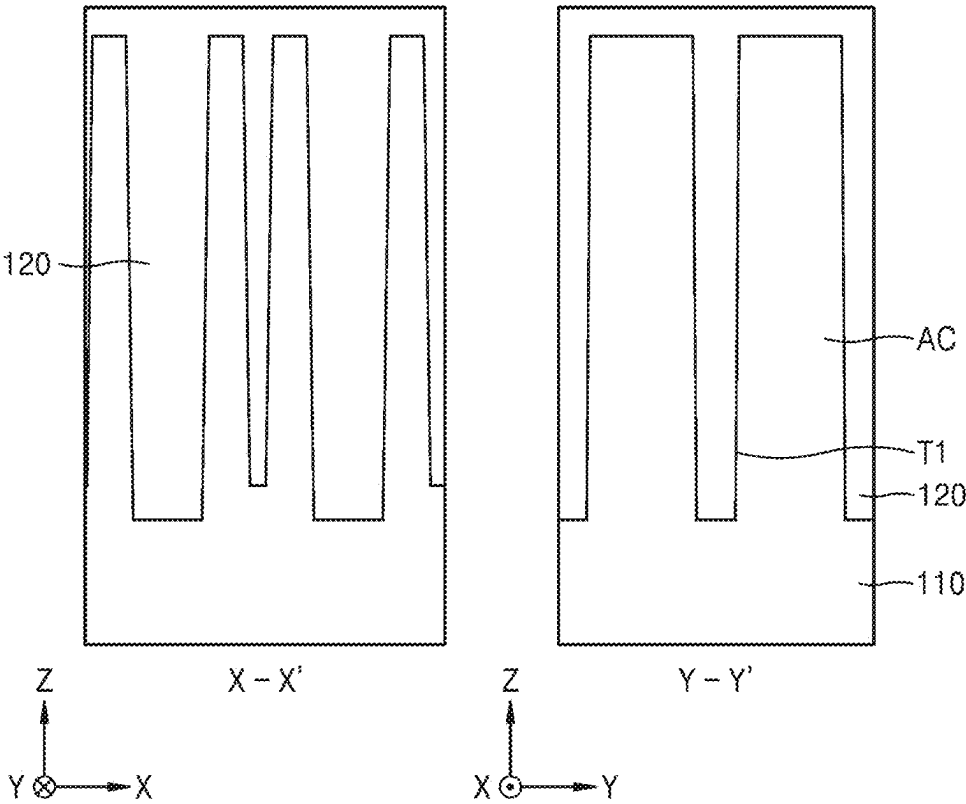


FIG. 7A

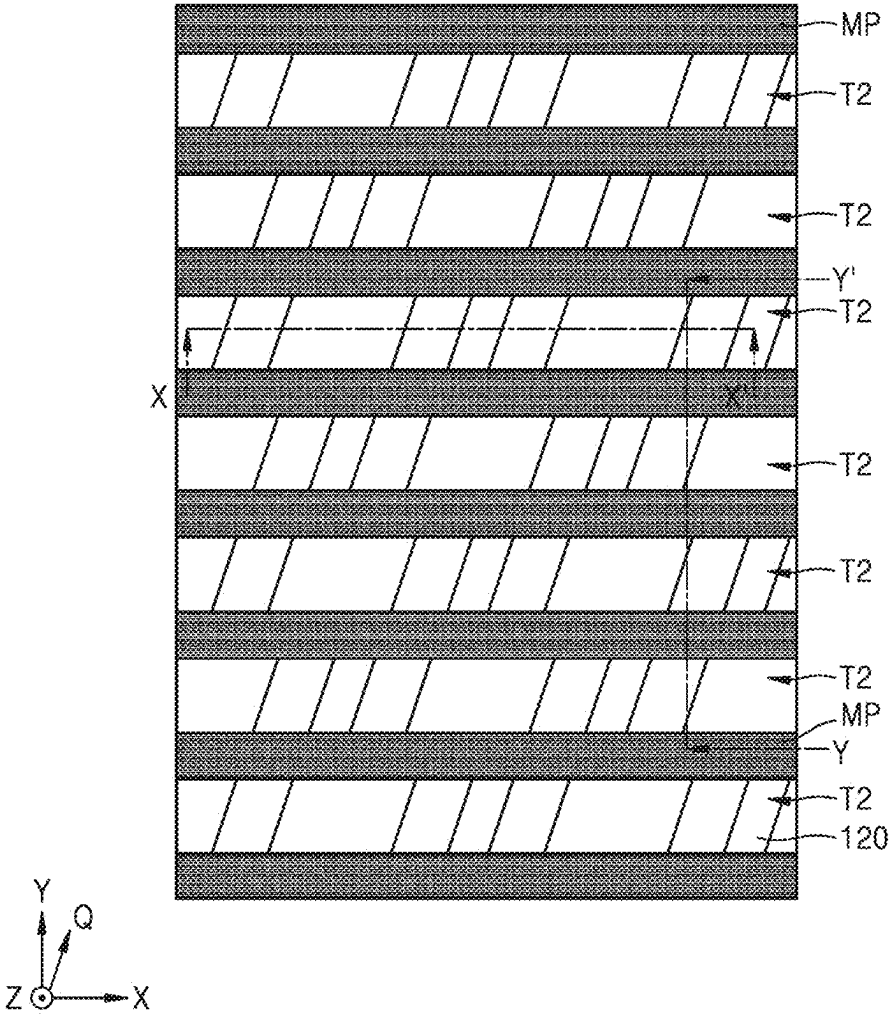


FIG. 7B

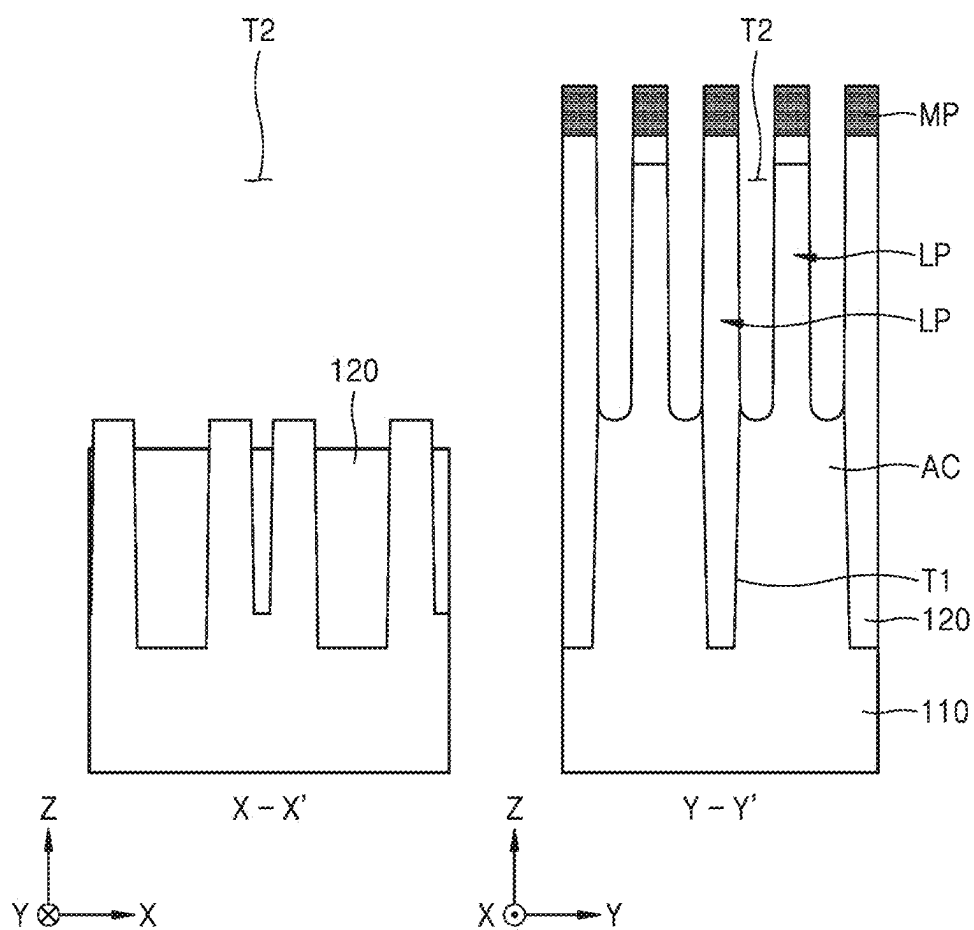


FIG. 8A

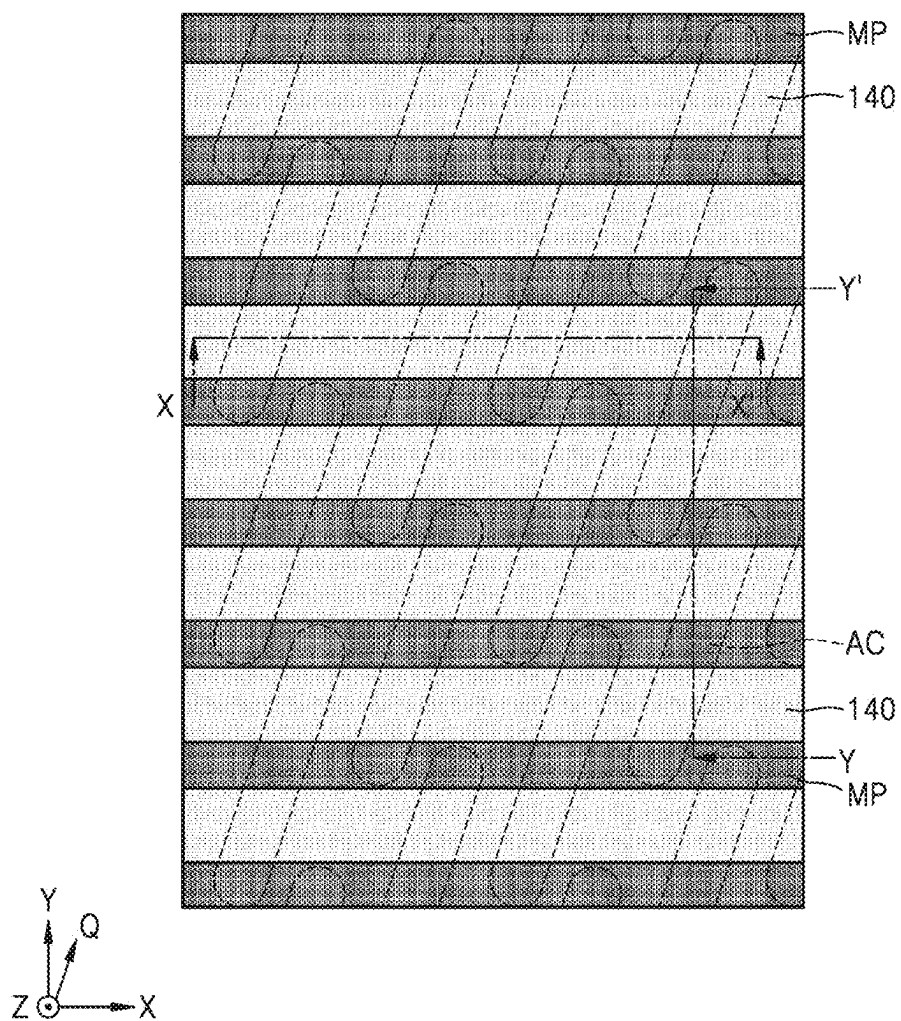


FIG. 8B

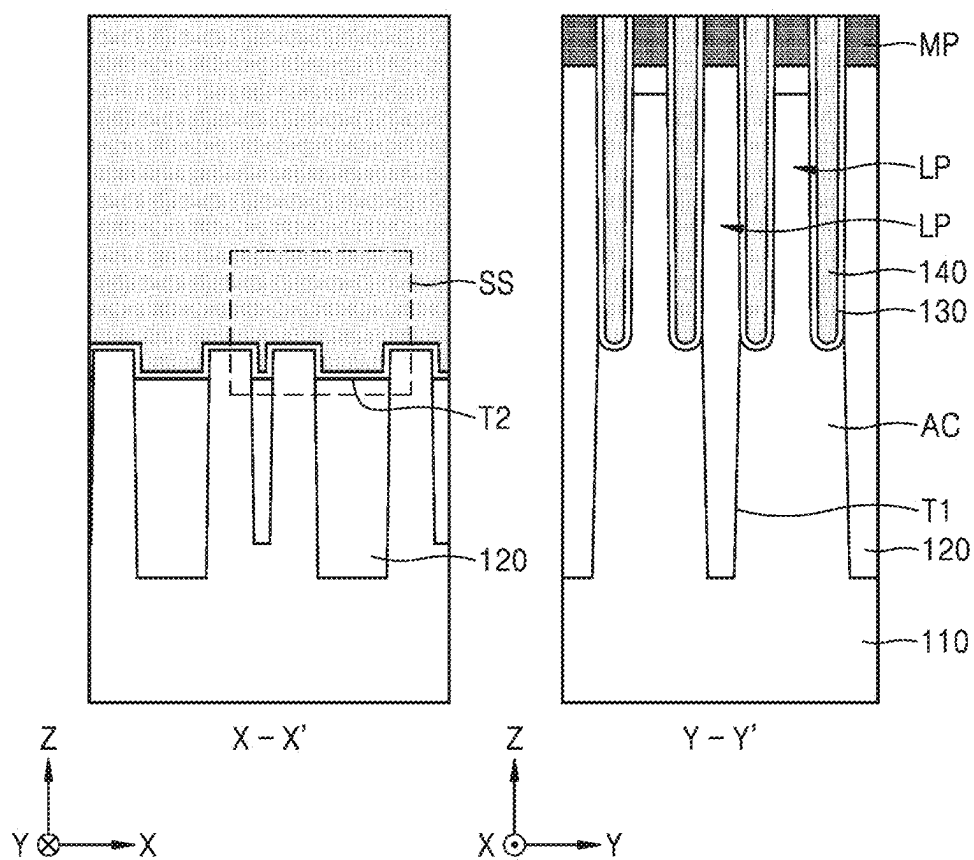


FIG. 9A

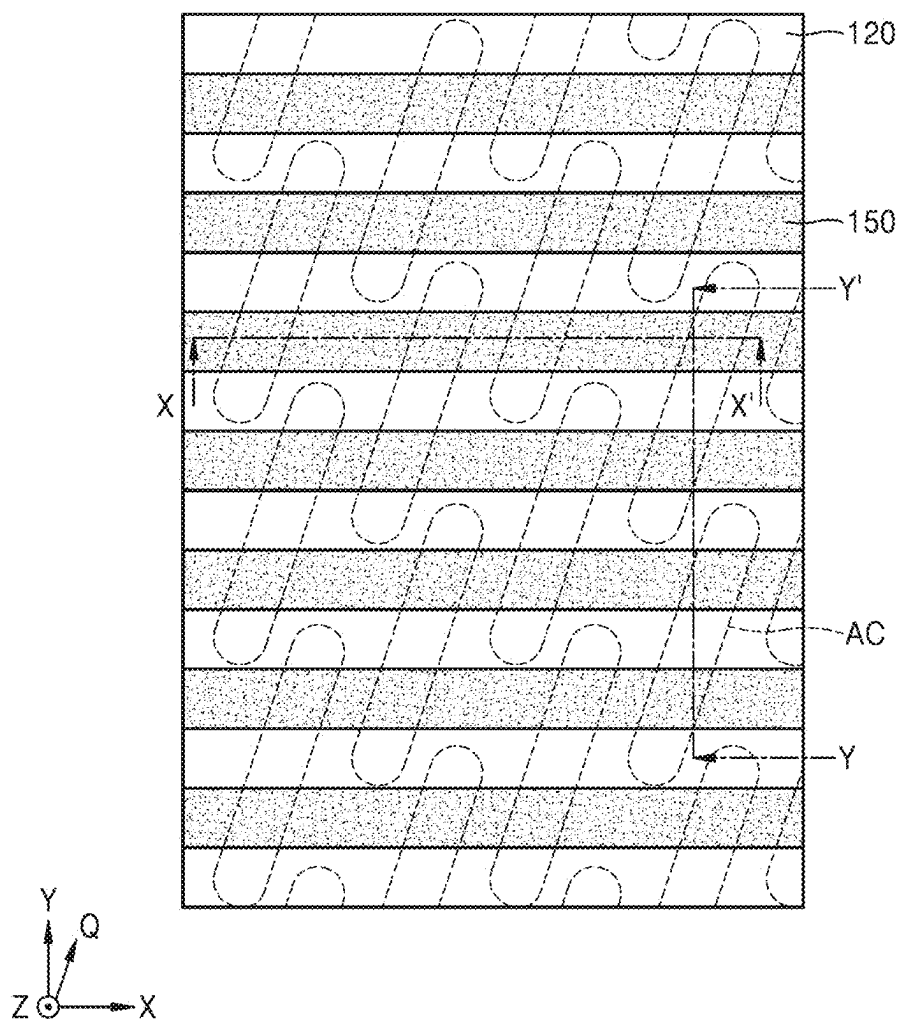


FIG. 9B

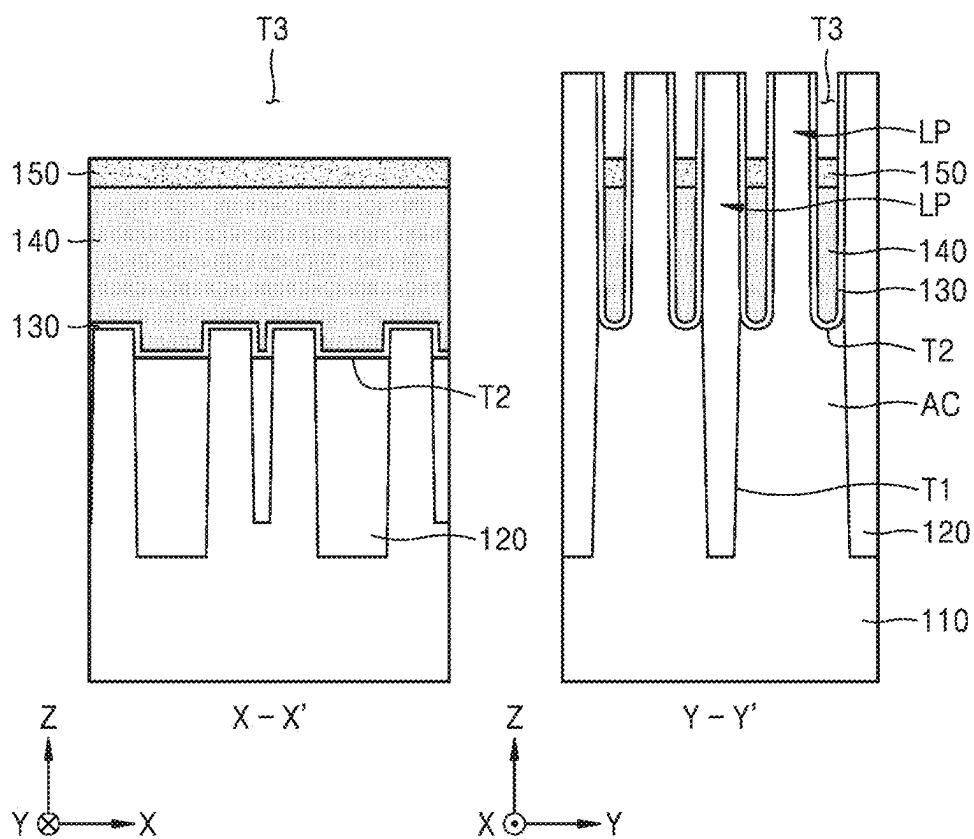


FIG. 10

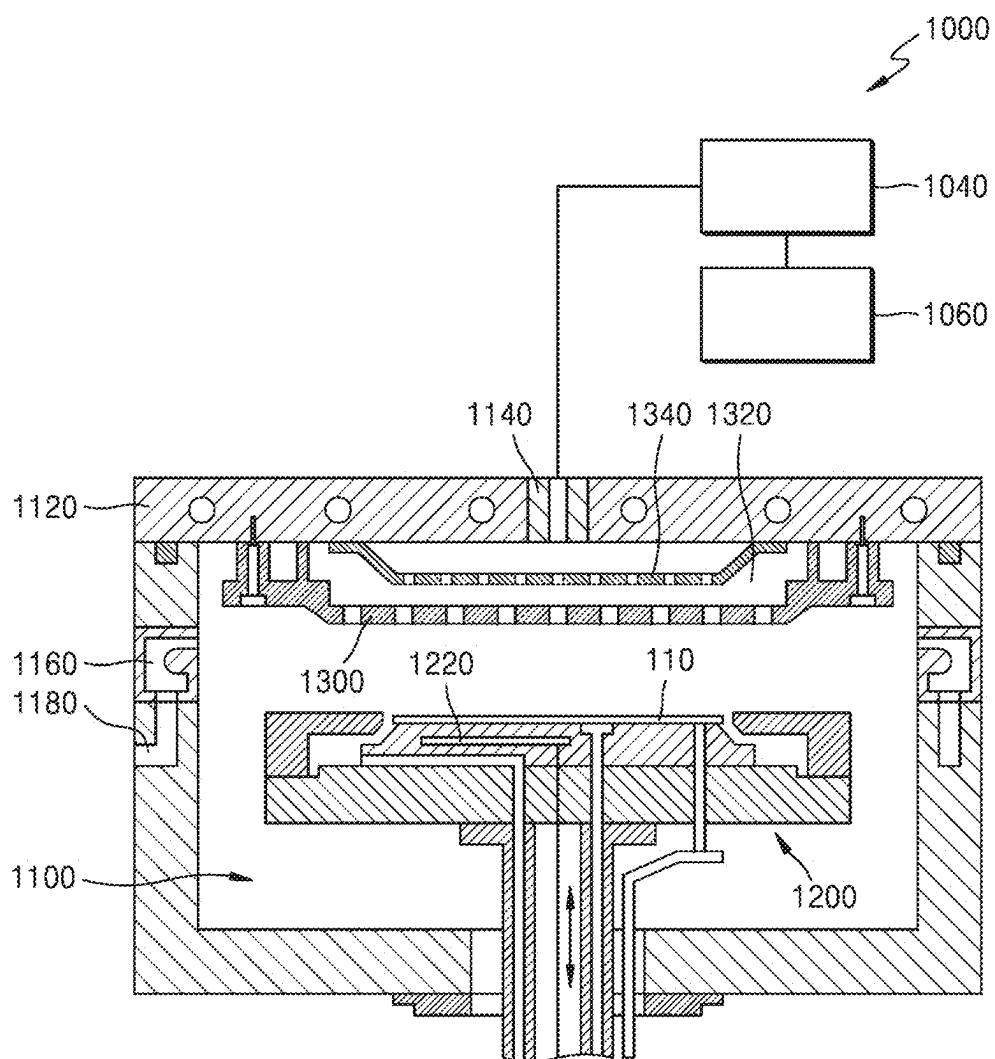
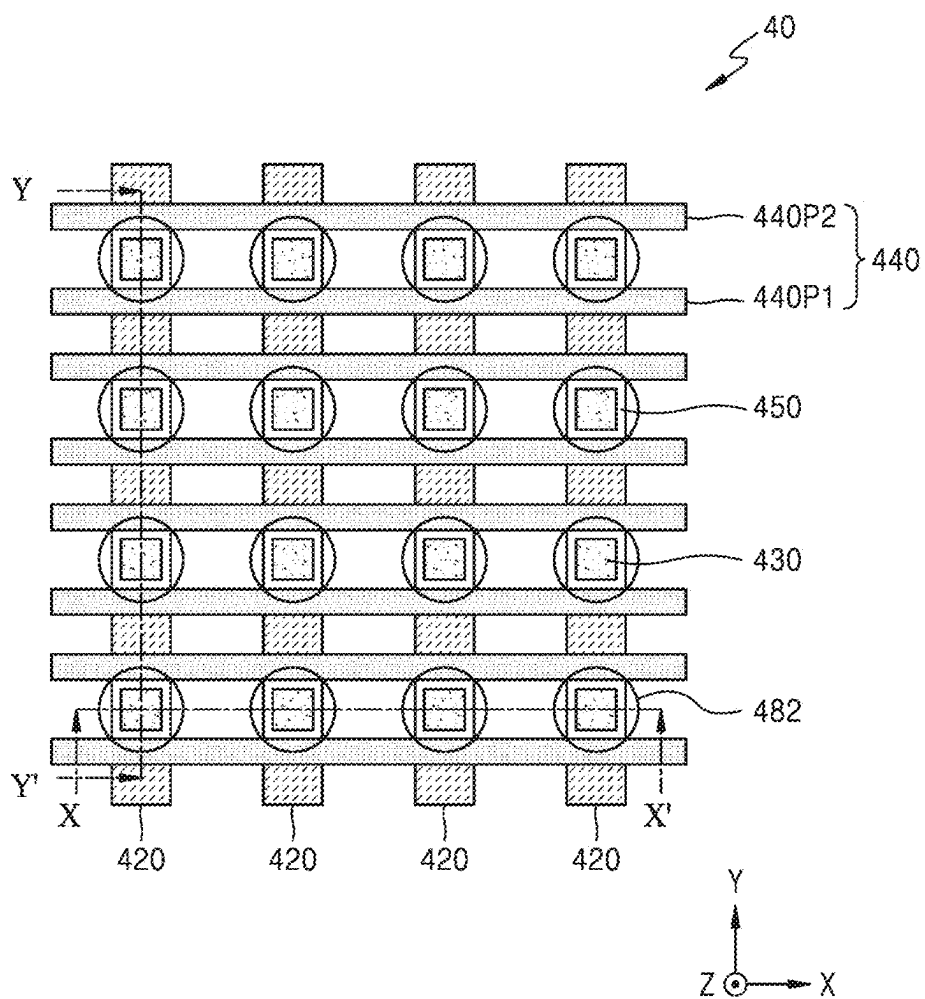


FIG. 11A



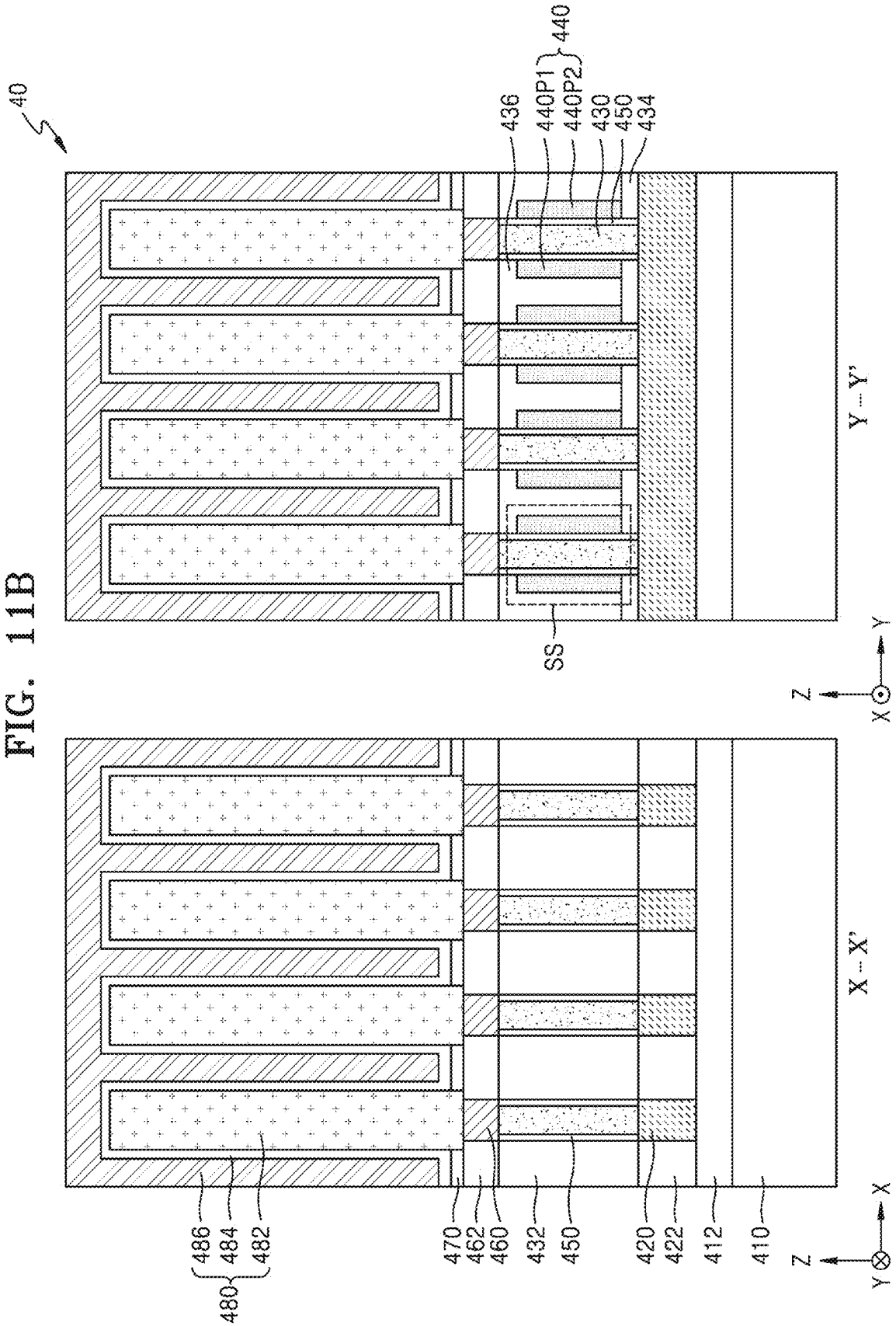


FIG. 11C

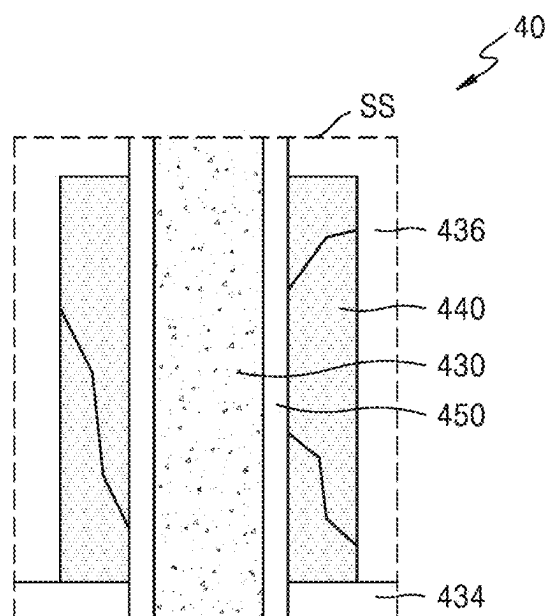


FIG. 12A

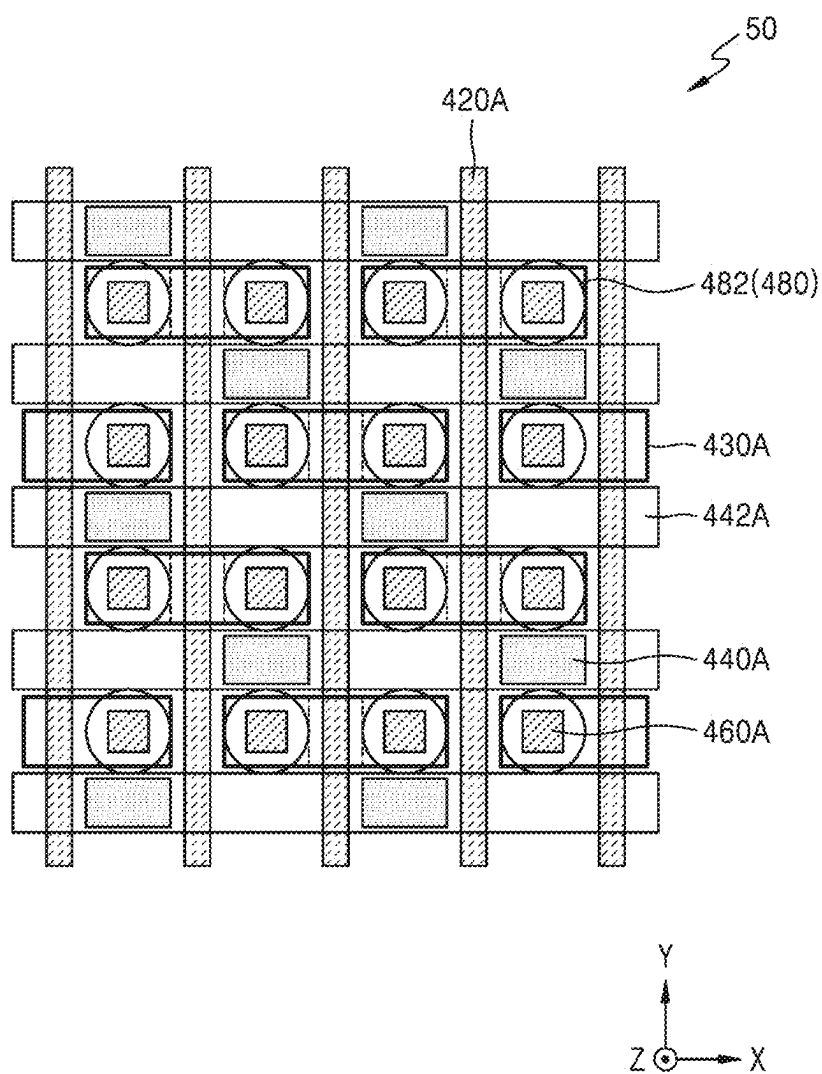
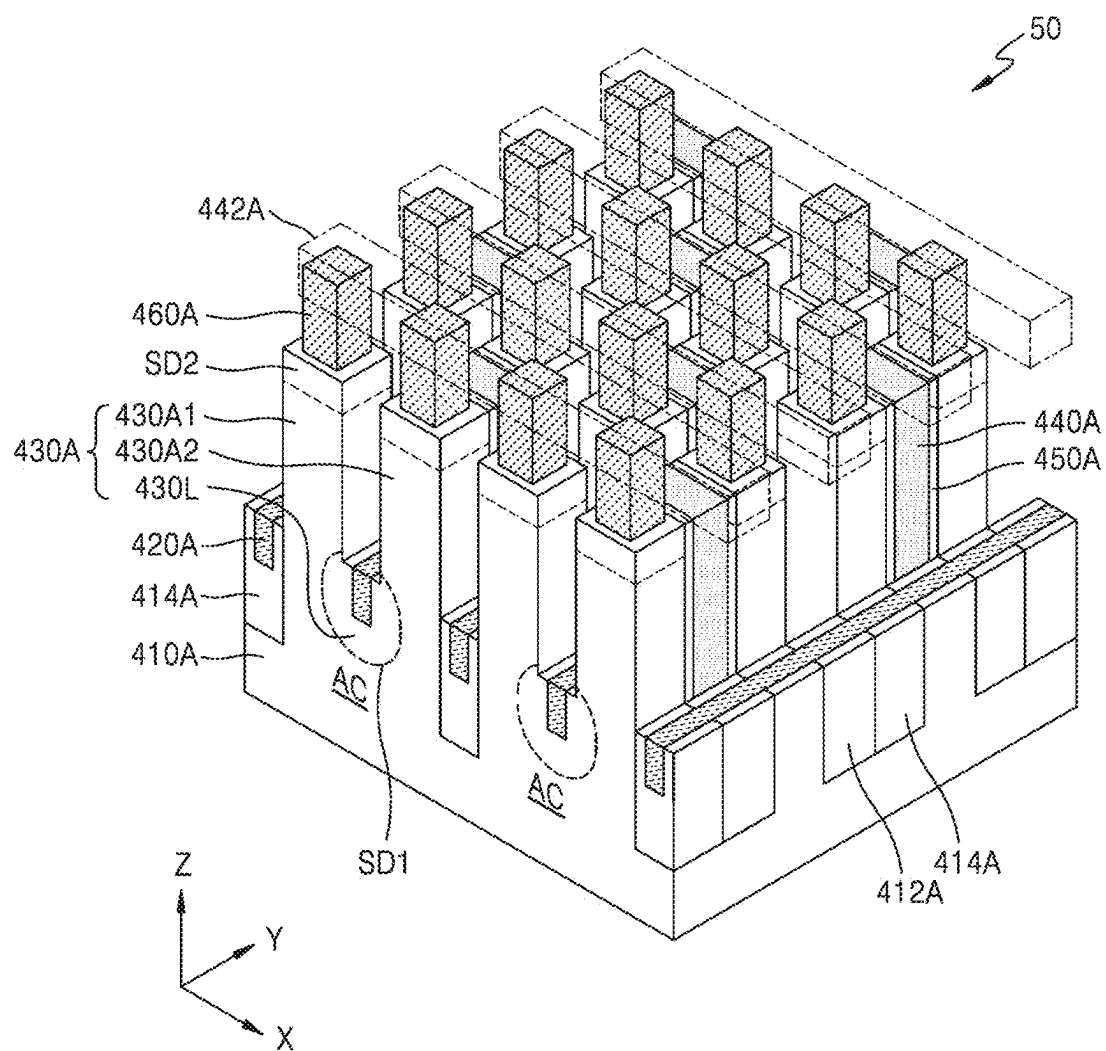


FIG. 12B



INTEGRATED CIRCUIT DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation of U.S. application Ser. No. 17/734,436, filed May 2, 2022, which is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0066825, filed on May 25, 2021, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] Inventive concepts relates to an integrated circuit device, and more particularly, to an integrated circuit device including a buried channel array transistor (BCAT).

[0003] Recently, as the degree of integration of an integrated circuit device has been gradually increased, a structure of an integrated circuit device including a BCAT having a plurality of word lines buried in a substrate has been proposed. Accordingly, various studies are being conducted to improve and stabilize an operation and reliability of the BCAT.

SUMMARY

[0004] Inventive concepts provides an integrated circuit device including a buried channel array transistor to improve electrical characteristics by adjusting a grain size of a metal material constituting a gate electrode.

[0005] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of example embodiments of the disclosure.

[0006] According to some example embodiments, an integrated circuit device includes a substrate including an active region defined by a device isolation layer, the substrate defining a gate trench extending across the active region, a gate dielectric layer conformally covering an inner surface of the gate trench, and a gate electrode filling the gate trench on the gate dielectric layer. The gate electrode is composed of crystal grains of a single metal, and a diagonal length of at least one of the crystal grains is greater than a height of the active region that is in contact with the gate electrode.

[0007] According to some example embodiments, an integrated circuit device includes a substrate including an active region defined by a device isolation layer, the substrate defining a gate trench extending across the active region, a gate dielectric layer conformally covering an inner surface of the gate trench, and a gate electrode filling the gate trench on the gate dielectric layer. The gate electrode is composed of crystal grains of a single metal, and the gate electrode comprises first crystal grains filling the gate trench, and second crystal grains between the first crystal grains and the gate dielectric layer, at least one of the second crystal grains smaller in grain size than at least one of the first crystal grains.

[0008] According to some example embodiments, an integrated circuit device includes a substrate including an active region defined by a device isolation layer, a gate structure crossing the active region and buried in the substrate, and source/drain regions on both sides of the gate structure. The gate structure is defined by a gate trench in the substrate, the gate structure including a gate dielectric layer conformal on

an inner wall of the gate trench, a gate barrier layer conformal on the gate dielectric layer in a lower region of the gate trench, a gate electrode on the gate barrier layer and filling the lower region of the gate trench, and a capping insulation layer on the gate electrode to fill an upper region of the gate trench. The gate electrode is composed of crystal grains of a single metal. The gate electrode comprises first crystal grains filling the gate trench, and second crystal grains between the first crystal grains and the gate dielectric layer, at least one of the second grains having a diagonal length less than a diagonal length of at least one of the first crystal grains. The diagonal length of the at least one of the first crystal grains is greater than a height a top surface of the active region exposed from the device isolation layer and a horizontal width of the top surface of the active region exposed from the device isolation layer.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Some example embodiments of inventive concepts will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings in which:

[0010] FIG. 1A is a plan view illustrating an integrated circuit device according to some example embodiments of inventive concepts, FIG. 1B is a cross-sectional view taken along line X-X' of FIG. 1A, and FIG. 1C is a cross-sectional view taken along line Y-Y' of FIG. 1A, and FIG. 1D is an enlarged cross-sectional view of a region SS of FIG. 1C;

[0011] FIG. 2 is an enlarged cross-sectional view illustrating an integrated circuit device according to some example embodiments of inventive concepts;

[0012] FIGS. 3A and 3B are cross-sectional views illustrating an integrated circuit device according to some example embodiments of inventive concepts, and FIG. 3C is an enlarged cross-sectional view illustrating a region SS of FIG. 3B;

[0013] FIG. 4 is a flowchart illustrating a process sequence of a method of fabricating an integrated circuit device, according to some example embodiments of inventive concepts;

[0014] FIGS. 5A, 6A, 7A, 8A, and 9A are plan views illustrating a process sequence of a method of fabricating an integrated circuit device, according to some example embodiments of inventive concepts, and FIGS. 5B, 6B, 7B, 8B, and 9B are cross-sectional views illustrating together cross-sections taken along lines X-X' and Y-Y' of FIGS. 5A, 6A, 7A, 8A, and 9A;

[0015] FIG. 10 is a cross-sectional configuration view of a process chamber that may be used to perform a method of fabricating an integrated circuit device according to some example embodiments of inventive concepts;

[0016] FIG. 11A is a layout illustrating an integrated circuit device according to some example embodiments of inventive concepts, FIG. 11B is a cross-sectional view illustrating together cross-sections taken along lines X-X' and Y-Y' of FIG. 11A, and FIG. 11C is an enlarged cross-sectional view illustrating a region SS of FIG. 11B; and

[0017] FIG. 12A is a layout illustrating an integrated circuit device according to some example embodiments of inventive concepts, and FIG. 12B is a perspective view illustrating the integrated circuit device of FIG. 12A.

DETAILED DESCRIPTION OF SOME EXAMPLE EMBODIMENTS

[0018] Hereinafter, some example embodiments of inventive concepts are described in detail with reference to the accompanying drawings.

[0019] FIG. 1A is a plan view illustrating an integrated circuit device according to some example embodiments of inventive concepts, FIG. 1B is a cross-sectional view taken along line X-X' of FIG. 1A, and FIG. 1C is a cross-sectional view taken along line Y-Y' of FIG. 1A, and FIG. 1D is an enlarged cross-sectional view of a region SS of FIG. 1C.

[0020] Referring to FIGS. 1A to 1D together, an integrated circuit device **10** includes a substrate **110** on which a plurality of active regions AC defined by a device isolation layer **120** and a plurality of gate trenches GT extending across the plurality of active regions AC are formed.

[0021] The plurality of active regions AC may be on or in the substrate **110**. The substrate **110** may include, for example, silicon (Si). Alternatively, the substrate **110** may include a semiconductor element such as germanium (Ge), or a compound semiconductor such as silicon carbide (SiC), gallium arsenide (GaAs), indium arsenide (InAs), or indium phosphide (InP). Alternatively, the substrate **110** may have a silicon on insulator (SOI) structure. For example, the substrate **110** may include a buried oxide layer (BOX). The substrate **110** may be in single-crystal phase or in polycrystalline phase. The substrate **110** may be doped, e.g. may be lightly doped with at least one of boron, phosphorus, or arsenic; however, example embodiments are not limited thereto.

[0022] The plurality of active regions AC may repeat to be spaced apart from each other in a first horizontal direction (X direction) and a second horizontal direction (Y direction) and may each have a shape extending in an oblique direction to have a long axis in one direction (Q direction) different from the first horizontal direction (X direction) and the second horizontal direction (Y direction). The plurality of gate trenches GT may have a plurality of line shapes extending parallel to each other in the first horizontal direction (X direction). The Q direction may intersect the X direction at an angle that is not ninety degrees; for example, the Q direction may intersect the X direction at an angle of forty-five degrees or more and 80 degrees or less; however, example embodiments are not limited thereto.

[0023] The device isolation layer **120** may define the plurality of active regions AC in the substrate **110**. The device isolation layer **120** may be formed of, e.g. may include, a silicon oxide, a silicon nitride, or a combination thereof. The device isolation layer **120** may extend in the first horizontal direction (X direction) on the substrate **110** and each may be spaced apart from each other in the second horizontal direction (Y direction) on the substrate **110**.

[0024] A gate dielectric layer **130**, a gate electrode **140** filling part of the gate trench GT on the gate dielectric layer **130**, a gate passivation layer **150** covering the gate electrode **140** in the gate trench GT, and a capping insulation layer **160** on the gate passivation layer **150** in the gate trench GT may be formed in each of the plurality of gate trenches GT.

[0025] The gate dielectric layer **130**, the gate electrode **140**, and the gate passivation layer **150** may constitute or correspond to a gate structure GS.

[0026] A level of an exposed portion of the substrate **110** on a bottom surface of the gate trench GT may be higher than a level of an exposed portion of the device isolation

layer **120**. Accordingly, a bottom surface of the gate structure GS may have a rough shape corresponding to a bottom profile of the plurality of gate trenches GT. For example, saddle-fin field effect transistors (saddle FINFETs) may be formed in the plurality of active regions AC.

[0027] The gate dielectric layer **130** covers an inner surface of the gate trench GT to be in contact with the active region AC and the device isolation layer **120**. In some example embodiments, the gate dielectric layer **130** may function as a gate insulating layer. In some example embodiments, the gate dielectric layer **130** may be formed with a thermal oxidation process such as with an in-situ steam generation (ISSG) process; however, example embodiments are not limited thereto.

[0028] The gate dielectric layer **130** may be formed of a silicon oxide, oxide/nitride/oxide (ONO), or a high-k dielectric layer having a higher dielectric constant than the silicon oxide. The high-k dielectric layer may include, for example, at least one of HfO₂, Al₂O₃, HfAlO₃, Ta₂O₃, TiO₂, or so on.

[0029] The gate electrode **140** may be formed of a metal, a metal nitride, a metal carbide, or a combination thereof. In some example embodiments, the gate electrode **140** may be formed of Ti, TiN, Ta, TaN, W, WN, TiSiN, WSiN, or a combination thereof. In the integrated circuit device **10** according to inventive concepts, the gate electrode **140** may be formed of a single metal, for example formed of a single tungsten (W) metal.

[0030] When the gate electrode **140** is formed of tungsten (W), crystal grains of the tungsten (W) may include or consist of, for example, ①, ②, ③, ④, ⑤, and ⑥ in the drawing, and each of the crystal grains may have a length from one vertex to another, e.g. may have a diagonal length (for example, **140D** of ⑤). In inventive concepts, the diagonal length of each crystal grain is used as a measure indicating a grain size of each of the crystal grains. The diagonal length of a crystal grain may be or correspond to a length from one vertex of the crystal grain to another vertex of the crystal grain, e.g. a largest length between vertices of the crystal grain. The diagonal length of a crystal grain may be measured in cross-section, e.g. in a cross-sectional direction as illustrated in FIG. 1D; however, example embodiments are not limited thereto. The diagonal length of a crystal grain may be determined by various analytic techniques such as but not limited to transmission electron microscopy (TEM) and/or x-ray diffraction (XRD); however, example embodiments are not limited thereto.

[0031] The number and shapes of the crystal grains ①, ②, ③, ④, ⑤, and ⑥ illustrated in the drawings are an example and not limited thereto. In addition, only the diagonal length **140D** of any one of the crystal grains ⑤ is illustratively described, but this may be equally applied to other crystal grains ①, ②, ③, ④, ⑤, and ⑥.

[0032] The diagonal length **140D** of each, a majority of, or at least one crystal grain may be greater than a height ACH of the active region AC that is in contact with the gate electrode **140**. The height ACH of the active region AC indicates a height of a portion at which the active region AC protrudes from the device isolation layer **120**. Alternatively or additionally, the diagonal length **140D** of each crystal grain may be greater than a horizontal width ACW of a top surface of the active region AC in contact with the gate electrode **140**.

[0033] Specifically, the diagonal lengths **140D** of crystal grains may be in a range of about 80 nm to about 100 nm but

are not limited thereto. Alternatively or additionally, each crystal grain may be configured to have a superlattice structure. In some example embodiments, in a cross-sectional view, the gate electrode **140** that is in contact with one active region AC may include seven or fewer grains of tungsten (W), e.g. averaged over various active regions AC. In the drawings, the number of crystal grains **①**, **②**, **③**, **④**, **⑤**, and **⑥** in contact with one active region AC is five but example embodiments are not limited thereto.

[0034] In order for each crystal grain to be formed in a size that is greater than or equal to a desired grain size, a fabrication facility (for example, a chemical vapor deposition facility) capable of forming a thin film with tungsten (W) may perform a process including the following recipe. In some example embodiments, each of or at least one of borane (BH_3), diborane (B_2H_6), monosilane (SiH_4), and tungsten hexafluoride (WF_6) may be used as a source gas of a process gas. Alternatively or additionally, a low-temperature process may include a nucleation layer forming process performed at a process temperature of about 250°C . or less and a bulk layer forming process performed at a process temperature of about 300°C . or less.

[0035] According to some examples, when the nucleation layer forming process is performed at a low-temperature process of about 250°C . or less, the reactivity of borane (BH_3) may be higher than the reactivity of diborane (B_2H_6). In this case, according to some example embodiments, by increasing (e.g. maximizing) dependence on borane (BH_3), sizes of crystal grains constituting a tungsten thin film may be relatively increased. Detailed description of the fabrication facility is described below with reference to FIG. 10. The increased size of the crystal grains included in the tungsten thin film may improve performance, such as speed and/or power, of the integrated circuit device **10**.

[0036] A gate passivation layer **150** may be on the gate electrode **140**. Specifically, the gate passivation layer **150** may be between the gate electrode **140** and the capping insulation layer **160**. The gate passivation layer **150** may prevent or reduce the impact form and/or probability of different materials forming the gate electrode **140** and the capping insulation layer **160** from mutually diffusing and/or reacting between the gate electrode **140** and the capping insulation layer **160**. The gate passivation layer **150** may be formed of or include, for example, doped or undoped polysilicon but is not limited thereto.

[0037] The capping insulation layer **160** fills the inside of the gate trench GT to be in contact with the gate dielectric layer **130**. The capping insulation layer **160** may be formed of a silicon nitride, a silicon oxynitride, or a combination thereof. For example, the capping insulation layer **160** may be composed of a second composition different from a first composition constituting the gate dielectric layer **130**. The second composition may include, for example, a silicon nitride.

[0038] The gate structure GS may have gate sidewalls facing the active region AC, and the capping insulation layer **160** has insulating sidewalls facing the active region AC. A source/drain region SD may be formed on both sides of the gate structure GS in the plurality of active regions AC. The source/drain region SD may be a raised source/drain region, e.g. may be formed with an epitaxial process; however, example embodiments are not limited thereto.

[0039] Recently, as the degree of integration of integrated circuit devices has been gradually increased, a structure of

the integrated circuit device **10** including a buried channel array transistor (BCAT) having a plurality of word lines buried in a substrate has been proposed. Accordingly, various studies have been conducted to improve and/or stabilize an operation of and/or reliability of the BCAT.

[0040] In general, an integrated circuit device includes a gate electrode composed of dense crystal grains (for example, having a diagonal length that is less than a height of an active region). According to inventive concepts, the integrated circuit device **10** may include the gate electrode **140** composed of crystal grains having a relatively large grain size.

[0041] The integrated circuit device **10** according to inventive concepts reduces most grain boundaries at a portion where crystal grains constituting the gate electrode **140** meet the active region AC, and thus, a phenomenon may be significantly improved in which characteristics of transistors are degraded due to electrical defects. For example, as a row disturb error and/or passing gate effects may be reduced.

[0042] According to inventive concepts, by adjusting a grain size of a metal material forming the gate electrode **140** in the integrated circuit device **10** including a BCAT, the electrical characteristics of the integrated circuit device **10** may be improved.

[0043] Ultimately, according to inventive concepts, an operation and/or reliability of the integrated circuit device **10** may be improved and stabilized. Further, although example embodiments are described with reference to an integrated circuit device **10** including a BCAT, example embodiments are not limited thereto.

[0044] FIG. 2 is an enlarged cross-sectional view illustrating an integrated circuit device according to some example embodiments of inventive concepts.

[0045] Most components constituting an integrated circuit device **20** to be described below and materials forming the components are substantially the same as or similar to the components and materials described above with reference to FIGS. 1A to 1D. Therefore, for the sake of convenient description, differences from the integrated circuit device **10** described above are mainly described.

[0046] Referring to FIG. 2, a gate electrode **240** of the integrated circuit device **20** may include first crystal grains and second crystal grains having a smaller grain size than the first crystal grains.

[0047] The gate electrode **240** of the integrated circuit device **20** according to the present embodiment may be formed of a single metal, preferably formed of a single tungsten (W) metal.

[0048] When the gate electrode **240** is formed of tungsten (W), crystal grains of the tungsten (W) may include relatively large first crystal grains (for example, **①**, **②**, **③**, **④**, **⑤**, and **⑥** in FIG. 2) and relatively small second crystal grains (for example, **a**, **b**, **c**, and **d** in FIG. 2). Each of the first crystal grains may have a first diagonal length (for example, **240D** of **⑤**).

[0049] The number and/or shapes of the first crystal grains **①**, **②**, **③**, **④**, **⑤**, and **⑥** and the number and shapes of the second crystal grains **a**, **b**, **c**, and **d** illustrated in the drawings are examples and not limited thereto. In addition, only the first diagonal length **240D** of the first crystal grain **5** of any one of the first crystal grains is

described as an example, but this may also be applied to the other first crystal grains ①, ②, ③, ④, ⑤, and ⑥ in the same manner.

[0050] The first diagonal length **240D** of the first crystal grains may be greater than the height **ACH** of the active region **AC** in contact with the gate electrode **240**. The height **ACH** of the active region **AC** indicates a height of a portion at which the active region **AC** protrudes from the device isolation layer **120**. Alternatively or additionally, the first diagonal length **240D** of the first crystal grains may be greater than a horizontal width **ACW** of a top surface of the active region **AC** in contact with the gate electrode **240**.

[0051] Specifically, the first diagonal lengths **240D** of the first crystal grains may be in a range of about 80 nm to about 100 nm but are not limited thereto. Alternatively or additionally second diagonal lengths (not illustrated) of the second crystal grains may be in a range of about 20 nm or less but are not limited thereto. Alternatively or additionally, each of the first crystal grains may be configured to have a superlattice structure, e.g. a single metal structure comprising of alternating layers of two or more materials; however, example embodiments are not limited thereto.

[0052] In order for each of the first crystal grains to be formed in a size that is greater than or equal to a desired grain size, a fabrication facility (for example, a chemical vapor deposition facility) capable of forming a thin film of tungsten (W) may perform a process with substantially the same recipe as the recipe described with reference to FIGS. 1A to 1D.

[0053] However, in this process, the second crystal grains having a relatively small size in at least a portion of an interface may be formed, e.g. due to a difference between a material (conductor) forming the gate electrode **240** and a material (insulator) forming the gate dielectric layer **130**.

[0054] FIGS. 3A and 3B are cross-sectional views of an integrated circuit device according to some example embodiments of inventive concepts, and FIG. 3C is an enlarged cross-sectional view illustrating a region **SS** of FIG. 3A.

[0055] Most components constituting an integrated circuit device **30** to be described below and materials forming the components are substantially the same as or similar to the components and materials described above with reference to FIGS. 1A to 1D. Therefore, for the sake of convenient description, differences from the integrated circuit device **10** described above are mainly described.

[0056] Referring to FIGS. 3A to 3C together, the integrated circuit device **30** may include a gate barrier layer **340** between the gate electrode **140** and the gate dielectric layer **130**.

[0057] In the integrated circuit device **30** according to some example embodiments, the gate barrier layer **340** may be formed to be in conformal contact with the gate dielectric layer **130**. Accordingly, the gate electrode **140** may be spaced apart from the gate dielectric layer **130**, and the gate electrode **140** may be surrounded by the gate barrier layer **340** and the gate passivation layer **150**.

[0058] In some example embodiments, the gate barrier layer **340** may be formed of a metal nitride, for example, at least one of TiN, TaN, WN, or so on but is not limited thereto. In the integrated circuit device **30** according to inventive concepts, the gate barrier layer **340** may be formed of a titanium nitride (TiN).

[0059] FIG. 4 is a flowchart illustrating a process sequence illustrating a method of fabricating an integrated circuit device according to some example embodiments of inventive concepts.

[0060] Referring to FIG. 4, a method **S10** of fabricating an integrated circuit device may include a process sequence of first to seventh steps **S110** to **S170**.

[0061] In a case in which example embodiment may be implemented differently, the process sequence may be performed differently from the sequence to be described below. For example, two processes described in succession may be performed at substantially the same time, e.g. on or within the same tool or equipment, or may be performed in a sequence opposite to the described sequence.

[0062] The method **S10** of fabricating an integrated circuit device according to inventive concepts may include the following steps but is not limited thereto.

[0063] The method **S10** of fabricating an integrated circuit device includes a first step (**S110**) of forming a device isolation trench defining an active region in a substrate, a second step **S120** of forming a device isolation layer filling the device isolation trench, a third step **S130** of forming a mask pattern on the active region and the device isolation layer, a fourth step **S140** of forming a space line by using the mask pattern as an etch mask, a fifth step **S150** of forming a gate dielectric layer and a gate electrode filling the space line, a sixth step **S160** of removing part of the gate electrode and forming a gate passivation layer, and a seventh step **S170** of forming a capping insulation layer covering a top surface of the gate passivation layer.

[0064] Technical characteristics of the first to seventh steps **S110** to **S170** are described in detail with reference to FIGS. 5A to 9B to be described below.

[0065] FIGS. 5A, 6A, 7A, 8A, and 9A are plan views illustrating a process sequence of a method of fabricating an integrated circuit device, according to some example embodiments of inventive concepts, and FIGS. 5B, 6B, 7B, 8B, and 9B are cross-sectional views illustrating together cross-sections taken along lines X-X' and Y-Y' of FIGS. 5A, 6A, 7A, 8A, and 9A.

[0066] Referring to FIGS. 5A and 5B together, a device isolation trench **T1** defining a plurality of active regions **AC** may be formed in the substrate **110**.

[0067] The plurality of active regions **AC** defined by the device isolation trench **T1** may be on the substrate **110**. The substrate **110** may be, for example, a semiconductor wafer including silicon (Si) such as lightly doped single-crystal silicon on a 200 mm, or 300 mm, or 450 mm wafer.

[0068] The plurality of active regions **AC** may repeat spaced apart from each other in the first horizontal direction (X direction) and the second horizontal direction (Y direction), and may each have a shape extending in an oblique direction to have a long axis in one direction (Q direction) different from the first horizontal direction (X direction) and the second horizontal direction (Y direction).

[0069] In the device isolation trench **T1**, a portion having a relatively great width may be formed to have a relatively great depth, and a portion having a relatively small width may be formed to have a relatively small depth.

[0070] Referring to FIGS. 6A and 6B together, the device isolation layer **120** filling the device isolation trench **T1** may be formed around the plurality of active regions **AC**.

[0071] The device isolation layer **120** may be formed to cover all the top surfaces of the plurality of active regions

AC. In a plan view, locations of the plurality of active regions AC are represented by dashed lines for better understanding. Portions of the device isolation layer **120** that cover the top surfaces of the plurality of active regions AC may protect or help to protect a surface of the substrate **110** during an ion injection process/ion implantation process for injecting impurity ions into the substrate **110** in subsequent processes and/or during a subsequent etching process.

[0072] The device isolation layer **120** may include a silicon oxide layer, a silicon nitride layer, or a combination thereof. At least one process selected from an atomic layer deposition (ALD) process, a chemical vapor deposition (CVD) process, a radical oxidation process, and a natural oxidation process may be used to form the device isolation layer **120**. The device isolation layer **120** may be formed with a shallow-trench isolation (STI) process and/or with a spin-on glass (SOG) process; however, example embodiments are not limited thereto.

[0073] Referring to FIGS. 7A and 7B together, mask patterns MP spaced apart from each other and extending parallel to each other in the first horizontal direction (X direction) may be formed on the plurality of active regions AC and the device isolation layer **120**.

[0074] The mask patterns MP may each be formed of a line-and-space pattern. For example, the mask patterns MP may each be formed of a silicon nitride, polysilicon, spin-on-hardmask (SOH), photoresist, or a combination thereof but are not limited thereto.

[0075] The plurality of active regions AC and the device isolation layer **120** are etched by using the mask patterns MP as an etch mask, and thus, a plurality of space lines T2 extending parallel to each other in the first horizontal direction (X direction) and a plurality of line patterns LP defining the plurality of space lines T2 may be formed.

[0076] The plurality of line patterns LP may include some of the plurality of active regions AC, part of the device isolation layer **120**, and the mask patterns MP.

[0077] Part of the substrate **110** and part of the device isolation layer **120** may be exposed on bottom surfaces of the plurality of space lines T2. As illustrated, on the bottom surfaces of the plurality of space lines T2, a level of the portion where the substrate **110** is exposed may be higher than a level of the portion where the device isolation layer **120** is exposed.

[0078] Referring to FIGS. 8A and 8B together, the gate dielectric layer **130** and the gate electrode **140** may be formed to fill the space line T2 between the line patterns LP.

[0079] The gate dielectric layer **130** in contact with a surface of the substrate **110** and a surface of the device isolation layer **120** in the space line T2, and the gate electrode **140** filling the remaining space of the space line T2 on the gate dielectric layer **130** may be formed.

[0080] On bottom surfaces of the plurality of space lines T2, a level of the portion where the substrate **110** is exposed may be higher than a level of the portion where the isolation layer **120** is exposed, and accordingly, a bottom surface of the gate electrode **140** may have a rough shape corresponding to a bottom profile of the plurality of space lines T2. Therefore, saddle FINFETs may be formed in the plurality of active regions AC.

[0081] The gate dielectric layer **130** may include an oxide layer formed by an atomic layer deposition (ALD) process and/or a thermal oxidation process such as an in-situ steam generation process. In some example embodiments,

the gate dielectric layer **130** may be formed of a silicon oxide, HfO_2 , Al_2O_3 , HfAlO_3 , Ta_2O_5 , TiO_2 , or oxide/nitride/oxide (ONO) but is not limited thereto.

[0082] Here, referring to a region SS of FIG. 8B corresponding to the region SS of FIG. 1D, the gate electrode **140** may be formed of, e.g. may include or consist of or consist essentially of, a single metal, for example a single tungsten (W) metal. When the gate electrode **140** is formed of tungsten (W), crystal grains of the tungsten (W) may include or consist of, for example, ①, ②, ③, ④, ⑤, and ⑥ in the drawing, and each of the crystal grains may have a diagonal length (for example, **140D** of ⑤).

[0083] The diagonal length **140D** of each crystal grain may be greater than a height ACH of the active region AC in contact with the gate electrode **140**. Alternatively or additionally, the diagonal length **140D** of each crystal grain may be greater than a horizontal width ACW of a top surface of the active region AC in contact with the gate electrode **140**.

[0084] Referring to FIGS. 9A and 9B together, an upper space T3 of the plurality of space lines is prepared by removing part of the gate electrode **140** by using the mask pattern MP (refer to FIG. 8B) as an etching mask, and then, part of the upper space T3 of the plurality of space lines may be filled with the gate passivation layer **150**.

[0085] The gate passivation layer **150** may include a silicon nitride layer and/or polysilicon. The gate passivation layer **150** may prevent or reduce the impact of and/or likelihood of different materials forming the gate electrode **140** and the capping insulation layer **160** (refer to FIG. 1B) to be formed in the subsequent process from mutually diffusing or reacting between the gate electrode **140** and the capping insulation layer **160**. In some example embodiments, the gate passivation layer **150** may be omitted.

[0086] Referring back to FIGS. 1A to 1D together, the capping insulation layer **160** may be formed to completely fill the upper space T3 of the plurality of space lines and to cover a top surface of the gate passivation layer **150**.

[0087] Next, a planarization process may be performed to remove an unnecessary material layer on the substrate **110** so that an upper surface **110T** of the substrate **110** is exposed. In some example embodiments, an etch-back process and/or a chemical mechanical polishing (CMP) process may be performed for the planarization process.

[0088] The integrated circuit device **10** according to inventive concepts may be fabricated by using the method S10 of fabricating an integrated circuit device described above. According to inventive concepts, by adjusting a grain size of a metal material forming the gate electrode **140** in the integrated circuit device **10** including a BCAT, the electrical characteristics of the integrated circuit device **10** are improved.

[0089] FIG. 10 is a cross-sectional configuration view of a process chamber that may be used to implement a method of fabricating an integrated circuit device according to some example embodiments of inventive concepts.

[0090] FIG. 10 schematically illustrates a process chamber **1000** of a chemical vapor deposition (CVD) facility that may be used to form a tungsten thin film on a substrate **110**.

[0091] The process chamber **1000** may be part of a fabrication facility including a plurality of chambers. The process chamber **1000** may include a sidewall, a bottom, and a cover **1120** forming a chamber interior region **1100**. The sidewall and bottom of the process chamber **1000** may be

made of a single aluminum block. The sidewall of the process chamber 1000 may include a conduit (not illustrated), and a fluid for adjusting a temperature of the sidewall of the process chamber 1000 may flow through the conduit. Alternatively or additionally, the process chamber 1000 may include a pumping ring 1160 connecting the chamber interior region 1100 to an exhaust port 1180.

[0092] A substrate support member 1200 capable of adjusting a temperature may be near the center of the chamber interior region 1100. The substrate support member 1200 supports the substrate 110 during a tungsten thin film forming process. In general, the substrate support member 1200 may be made of aluminum, ceramic, or a combination of aluminum and ceramic and may include a vacuum port (not illustrated) and one or more heating members 1220.

[0093] Vacuum may be applied between the substrate 110 and the substrate support member 1200 by using the vacuum port to fix the substrate 110 to the substrate support member 1200 during the tungsten thin film forming process. The heating member 1220 may be in the substrate support member 1200 to heat the substrate support member 1200 and the substrate 110 on the substrate support member 1200 to a certain temperature.

[0094] The cover 1120 may be supported by the sidewall of the process chamber 1000 and removed for maintenance of the process chamber 1000. In general, the cover 1120 may be made of aluminum and may have a conduit therein, and a fluid for adjusting a temperature of the cover 1120 may flow through the conduit.

[0095] A mixing block 1140 may be inside the cover 1120. The mixing block 1140 may be connected to a gas supply source 1040. In general, individual gases supplied from the gas supply source 1040 are mixed with each other in the mixing block 1140. The gases are mixed into a single uniform gas flow in the mixing block 1140, and the single uniform gas flow is supplied to the chamber interior region 1100 through a shower head 1300. The gas supply source 1040 may be located at or near other components of the process chamber 1000, or may be located in a separate facility from the process chamber 1000.

[0096] The shower head 1300 may be connected to the cover 1120. In addition, a perforated blocker plate 1340 may be selectively arranged in a shower head interior region 1320 between the shower head 1300 and the cover 1120. A gas to be supplied to the chamber interior region 1100 through the mixing block 1140 is first diffused by the perforated blocker plate 1340. Then, the gas is supplied to the chamber interior region 1100 through the shower head 1300. The perforated blocker plate 1340 and the shower head 1300 are configured to provide a uniform gas flow into the chamber interior region 1100. The uniform gas flow promotes formation of a uniform tungsten thin film on the substrate 110.

[0097] A gas line for supplying process gases such as a tungsten-containing gas and a reducing gas from the gas supply source 1040 to the chamber interior region 1100 may include a valve (not illustrated) for diverting a gas flow.

[0098] Alternatively or additionally, the gas supply source 1040 may be controlled by a gas controller 1060. For example, the gas controller 1060 may control at least one of the type of gas supplied to the chamber interior region 1100, a supply time and an end time of the gas, and a flow rate of the gas by controlling the gas supply source 1040. The gas controller 1060 may be a mass flow controller, and/or a

programmable logic controller (PLC); however, example embodiments are not limited thereto. The gas controller 1060 may include or be implemented in processing circuitry such as hardware including logic circuits; a hardware/software combination such as a processor executing software; or a combination thereof. For example, the processing circuitry more specifically may include, but is not limited to, a central processing unit (CPU), an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FPGA), a System-on-Chip (SoC), a programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), etc. The processing circuitry may include electrical components such as at least one of transistors, resistors, capacitors, etc. The processing circuitry may include electrical components such as logic gates including at least one of AND gates, OR gates, NAND gates, NOT gates, etc.

[0099] In order to fabricate the integrated circuit devices 10, 20, and 30 according to inventive concepts described above, the gas supply source 1040 may supply borane (BH_3), diborane (B_2H_6), monosilane (SiH_4), and tungsten hexafluoride (WF_6), at least one of which may be a process gas, and the gas controller 1060 may control the process gas to an appropriate value.

[0100] Some example embodiments according to inventive concepts in which a gas is supplied into the process chamber 1000 to form a tungsten thin film is described as follows.

[0101] First, the substrate 110 is loaded into the process chamber 1000, and the substrate 110 is mounted on the substrate support member 1200. The substrate 110 may include a plurality of rough patterns formed by the active region AC (refer to FIG. 8B), and the gate dielectric layer 130 formed on the plurality of rough patterns.

[0102] Next, a certain amount of process gases is supplied from the gas supply source 1040 to the mixing block 1140 under the control of the gas controller 1060, and the process gas is substantially and uniformly supplied from the shower head 1300 to the chamber interior region 1100.

[0103] At the same time, the heating member 1220 located in the substrate support member 1200 is driven to emit thermal energy while maintaining the chamber internal region 1100 at a certain pressure by exhausting an atmosphere of the chamber interior region 1100 to the exhaust port 1180.

[0104] The emitted thermal energy may heat an upper portion of the substrate support member 1200 to heat the substrate 110 mounted on the substrate support member 1200 to a certain temperature. The supplied process gas may cause a chemical reaction to form a tungsten thin film on the entire surface of the substrate 110.

[0105] In order to fabricate the integrated circuit devices 10, 20, and 30 according to inventive concepts described above, the heating member 1220 may operate such that a low-temperature process including a nucleation layer forming process performed at a process temperature of about 250° C. or less and a bulk layer forming process performed at a process temperature of about 300° C. or less is performed.

[0106] FIG. 11A is a layout illustrating an integrated circuit device according to some example embodiments of inventive concepts, FIG. 11B is a cross-sectional view illustrating together cross-sections taken along lines X-X'

and Y-Y' of FIG. 11A, and FIG. 11C is an enlarged cross-sectional view illustrating a region SS of FIG. 11B.

[0107] Referring to FIGS. 11A to 11C, an integrated circuit device 40 includes a substrate 410, a plurality of first conductive lines 420, a channel layer 430, a gate electrode 440, and a gate insulating layer 450, and a capacitor structure 480.

[0108] The integrated circuit device 40 may include a memory device including a vertical channel transistor (VCT). The VCT may refer to a structure in which a channel length of the channel layer 430 extends in a vertical direction from the substrate 410.

[0109] A lower insulating layer 412 may be on the substrate 410, and a plurality of first conductive lines 420 may be on the lower insulating layer 412 to be spaced apart from each other in the first horizontal direction (X direction) and to extend in the second horizontal direction (Y direction). A plurality of first insulation patterns 422 may be on the lower insulating layer 412 to fill a space between the plurality of first conductive lines 420. The plurality of first insulation patterns 422 may extend in the second horizontal direction (Y direction), and an upper surface of the plurality of first insulation patterns 422 may be at the same level as an upper surface of the plurality of first conductive lines 420. The plurality of first conductive lines 420 may function as bit lines of the integrated circuit device 40; however, example embodiments are not limited thereto.

[0110] In some example embodiments, the plurality of first conductive lines 420 may include doped polysilicon, a metal, a conductive metal nitride, a conductive metal silicide, a conductive metal oxide, or a combination thereof. For example, the plurality of first conductive lines 420 may be formed of doped polysilicon, Al, Cu, Ti, Ta, Ru, W, Mo, Pt, Ni, Co, TiN, TaN, WN, NbN, TiAl, TiAlN, TiSi, TiSiN, TaSi, TaSiN, RuTiN, NiSi, CoSi, IrOx, RuOx, or a combination thereof, but example embodiments are not limited thereto. The plurality of first conductive lines 420 may include a single layer or multiple layers of the above-described materials. In some example embodiments, the plurality of first conductive lines 420 may include a two-dimensional (2D) semiconductor material, and for example, the 2D semiconductor material may include graphene, carbon nanotube, or a combination thereof.

[0111] Channel layers 430 may be arranged in a matrix in which the channel layers 430 are arranged on the plurality of first conductive lines 420 to be spaced apart from each other in the first horizontal direction (X direction) and the second horizontal direction (Y direction). The channel layer 430 may have a first width in the first horizontal direction (X direction) and have a first height in a vertical direction (Z direction), and the first height may be greater than the first width. For example, the first height may be about 2 times to about 10 times the first width but is not limited thereto. A bottom portion of the channel layer 430 may function as a first source/drain region (not illustrated), an upper portion of the channel layer 430 may function as a second source/drain region (not illustrated), and a portion of the channel layer 430 between first and second source/drain regions may function as a channel region (not illustrated).

[0112] In some example embodiments, the channel layer 430 may include an oxide semiconductor, for example, the oxide semiconductor may include $\text{In}_x\text{Ga}_y\text{Zn}_z\text{O}$, $\text{In}_x\text{Ga}_y\text{Si}_z\text{O}$, $\text{In}_x\text{Sn}_y\text{Zn}_z\text{O}$, $\text{In}_x\text{Zn}_y\text{O}$, Zn_xO , $\text{Zn}_x\text{Sn}_y\text{O}$, $\text{Zn}_x\text{O}_y\text{N}$, $\text{Zr}_x\text{Zn}_y\text{Sn}_z\text{O}$, Sn_xO , $\text{Hf}_x\text{In}_y\text{Zn}_z\text{O}$, $\text{Ga}_x\text{Zn}_y\text{Sn}_z\text{O}$, $\text{Al}_x\text{Zn}_y\text{Sn}_z\text{O}$,

$\text{Yb}_x\text{Ga}_y\text{Zn}_z\text{O}$, $\text{In}_x\text{Ga}_y\text{O}$, or a combination thereof. The channel layer 430 may include a single layer or layers of the oxide semiconductor. In some examples embodiments the channel layer 430 may have a bandgap energy that is greater than a bandgap energy of silicon. For example, the channel layer 430 may have a bandgap energy of about 1.5 eV to about 5.6 eV. For example, the channel layer 430 may have the best channel performance when having a bandgap energy of about 2.0 eV to about 4.0 eV. For example, the channel layer 430 may be polycrystalline or amorphous but is not limited thereto. In some example embodiments, the channel layer 430 may include a 2D semiconductor material, and for example, the 2D semiconductor material may include graphene, carbon nanotubes, or a combination thereof.

[0113] The gate electrode 440 may extend in the first horizontal direction (X direction) on both sidewalls of the channel layer 430. The gate electrode 440 may include a first sub-gate electrode 440P1 facing a first sidewall of the channel layer 430 and a second sub-gate electrode 440P2 facing a second sidewall opposite to the first sidewall of the channel layer 430. Because one channel layer 430 is between the first sub-gate electrode 440P1 and the second sub-gate electrode 440P2, the integrated circuit device 40 may have a dual-gate transistor structure. However, inventive concepts are not limited thereto, and a single gate transistor structure may also be implemented by omitting the second sub-gate electrode 440P2 and by forming only the first sub-gate electrode 440P1 facing the first sidewall of the channel layer 430.

[0114] The gate electrode 440 may be formed of a metal, a metal nitride, a metal carbide, or a combination thereof. In some example embodiments, the gate electrode 440 may be formed of Ti, TiN, Ta, TaN, W, WN, TiSiN, WSiN, or a combination thereof. In the integrated circuit device 40 according to inventive concepts, the gate electrode 440 may be formed of a single metal, preferably a single tungsten (W) metal.

[0115] When the gate electrode 440 is formed of tungsten (W), crystal grains of tungsten (W) may have a relatively large grain size. The number and/or shapes of the crystal grains illustrated in the drawings are examples and not limited thereto.

[0116] Specifically, diagonal lengths of crystal grains constituting the gate electrode 440 may be in a range of about 80 nm to about 100 nm but are not limited thereto. Alternatively or additionally, each of the crystal grains may be configured to have a superlattice structure.

[0117] In order for each crystal grain to be formed in a size that is greater than or equal to a desired grain size, a fabrication facility (for example, a chemical vapor deposition facility) capable of forming a thin film with tungsten (W) may perform a process with a recipe similar to, e.g. substantially the same as, the recipe described with reference to FIGS. 1A to 1D.

[0118] The gate insulating layer 450 may surround a sidewall of the channel layer 430 and may be between the channel layer 430 and the gate electrode 440. For example, the entire sidewall of the channel layer 430 may be surrounded by the gate insulating layer 450, and part of the sidewall of the gate electrode 440 may be in contact with the gate insulating layer 450. In other embodiments, the gate insulating layer 450 may extend in an extension direction (that is, the first horizontal direction (X direction)) of the

gate electrode **440**, and among sidewalls of the channel layer **430**, only two sidewalls facing the gate electrode **440** may also be in contact with the gate insulating layer **450**.

[0119] In some example embodiments, the gate insulating layer **450** may include a silicon oxide film, a silicon oxynitride film, a high-k dielectric layer with a higher dielectric constant than a dielectric constant of the silicon oxide film, or a combination thereof. The high-k dielectric layer may be formed of a metal oxide or a metal oxynitride. For example, the high-k dielectric layer capable of being used as the gate insulating layer **450** may be formed of HfO_2 , HfSiO , HfSiON , HfTaO , HfTiO , HfZrO , ZrO_2 , Al_2O_3 , or a combination thereof but is not limited thereto.

[0120] A plurality of second insulation patterns **432** may extend in the second horizontal direction (Y direction) on the plurality of first insulation patterns **422**, and the channel layer **430** may be between two adjacent second insulation patterns **432** of the plurality of second insulation patterns **432**. In addition, a first buried layer **434** and a second buried layer **436** may be in a space between two adjacent channel layers **430**, between two adjacent second insulation patterns **432**. The first buried layer **434** may be at a bottom portion of the space between two adjacent channel layers **430**, and the second buried layer **436** may be on the first buried layer **434** to fill the rest of the space between the two adjacent channel layers **430**. A top surface of the second buried layer **436** may be at the same level as a top surface of the channel layer **430**, and the second buried layer **436** may cover the top surface of the gate electrode **440**. Alternatively or additionally, the plurality of second insulation patterns **432** may also be formed as material layers continuous to the plurality of first insulation patterns **422**, or the second buried layer **436** may also be formed as a continuous material layer to the first buried layer **434**.

[0121] Structures such as capacitor contacts **460** may be on the channel layers **430**. The capacitor contacts **460** may vertically overlap with the channel layers **430** and may be arranged in a matrix in which the capacitor contacts **460** are spaced apart from each other in the first horizontal direction (X direction) and the second horizontal direction (Y direction). The capacitor contacts **460** may each be formed of doped polysilicon, Al, Cu, Ti, Ta, Ru, W, Mo, Pt, Ni, Co, TiN, TaN, WN, NON, TiAl, TiAlN, TiSi, TiSiN, TaSi, TaSiN, RuTiN, NiSi, CoSi, IrOx, RuOx, or a combination thereof but are not limited thereto. The upper insulating layer **462** may surround sidewalls of the capacitor contact **460** on the plurality of second insulation patterns **432** and the second buried layer **436**.

[0122] An etch stop layer **470** may be on the upper insulating layer **462**, and the capacitor structure **480** may be on the etch stop layer **470**. The capacitor structure **480** may include a lower electrode **482**, a capacitor dielectric layer **484**, and an upper electrode **486**.

[0123] The lower electrode **482** may pass through the etch stop layer **470** to be electrically connected to an upper surface of the capacitor contact **460**. The lower electrode **482** may be formed in a pillar type extending in the vertical direction (Z direction) but is not limited thereto. In some example embodiments, the lower electrode **482** may be arranged to vertically overlap the capacitor contact **460** and may be arranged in a matrix in which the lower electrodes are spaced apart from each other in the first horizontal direction (X direction) and the second horizontal direction (Y direction). In addition, a landing pad (not illustrated) may

also be between the capacitor contact **460** and the lower electrode **482** such that the lower electrode **482** may have a hexagonal shape.

[0124] FIG. 12A is a layout illustrating an integrated circuit device according to some example embodiments of inventive concepts, and FIG. 12B is a perspective view illustrating the integrated circuit device of FIG. 12A.

[0125] Referring to FIGS. 12A and 12B together, an integrated circuit device **50** may include a substrate **410A**, a plurality of first conductive lines **420A**, a channel structure **430A**, a contact gate electrode **440A**, a plurality of second conductive lines **442A**, and a capacitor structure **480**. The integrated circuit device **50** may be or include a memory device including a VCT.

[0126] A plurality of active regions AC may be defined in the substrate **410A** by the first device isolation layer **412A** and the second device isolation layer **414A**. The channel structure **430A** may be in each of the active regions AC, and the channel structure **430A** may include a first active pillar **430A1** and a second active pillar **430A2** respectively extending in a vertical direction, and a connection portion **430L** connected to a bottom of the active pillar **430A1** and a bottom of the second active pillar **430A2**. A first source/drain region SD1 may be in the connection portion **430L**, and second source/drain regions SD2 may be formed above the first and second active pillars **430A1** and **430A2**. The first and second active pillars **430A1** and **430A2** may each constitute an independent unit memory cell.

[0127] The plurality of first conductive lines **420A** may extend in a direction crossing each of the plurality of active regions AC, for example, in the second horizontal direction (Y direction). One of the plurality of first conductive lines **420A** may be on the connection portion **430L** between the first and second active pillars **430A1** and **430A2**, and the one first conductive line **420A** may be on the first source/drain region SD1. Another first conductive line **420A** adjacent to the one first conductive line **420A** may be between the two channel structures **430A**. One of the plurality of first conductive lines **420A** may function as a common bit line included in two unit memory cells constituting the first and second active pillars **430A1** and **430A2** arranged on both sides of the one first conductive line **420A**.

[0128] One contact gate electrode **440A** may be between two adjacent channel structures **430A** in the second horizontal direction (Y direction). For example, the contact gate electrode **440A** may be between the first active pillar **430A1** included in one channel structure **430A** and the second active pillar **430A2** of the channel structure **430A** adjacent thereto, and one contact gate electrode **440A** may be shared by the first and second active pillars **430A1** and **430A2** arranged on both the sidewalls of the contact gate electrode **440A**. A gate insulating layer **450A** may be between the contact gate electrode **440A** and the first active pillar **430A1** and between the contact gate electrode **440A** and the second active pillar **430A2**. The plurality of second conductive lines **442A** may extend in the first horizontal direction (X direction) on an upper surface of the contact gate electrode **440A**. The plurality of second conductive lines **442A** may function as word lines of the integrated circuit device **50**.

[0129] The contact gate electrode **440A** may be formed of, include, consist of, or consist essentially of a metal, a metal nitride, a metal carbide, or a combination thereof. In some embodiments, the contact gate electrode **440A** may be formed of Ti, TiN, Ta, TaN, W, WN, TiSiN, WSiN, or a

combination thereof. In the integrated circuit device **40** according to inventive concepts, the contact gate electrode **440A** may be formed of a single metal, preferably, a single tungsten (W) metal.

[0130] When the contact gate electrode **440A** is formed of tungsten (W), crystal grains of tungsten (W) may have a relatively large grain size. The number and shapes of the crystal grains illustrated in the drawings are examples and not limited thereto.

[0131] Specifically, diagonal lengths of crystal grains constituting the contact gate electrode **440A** may be in a range of about 80 nm to about 100 nm but are not limited thereto. Alternatively or additionally, each of the crystal grains may be configured to have a superlattice structure.

[0132] In order for each of crystal grains to be formed in a size that is greater than or equal to a desired grain size, a fabrication facility (for example, a chemical vapor deposition facility) capable of forming a thin film of tungsten (W) may perform a process with substantially the same recipe as the recipe described with reference to FIGS. 1A to 1D.

[0133] A capacitor contact **460A** may be on the channel structure **430A**. The capacitor contact **460A** may be on the second source/drain region **SD2**, and the capacitor structure **480** may be on the capacitor contact **460A**.

[0134] While inventive concepts has been particularly shown and described with reference to some example embodiments thereof, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the following claims. Furthermore none of the example embodiments are necessarily mutually exclusive with one another. For example some example embodiments may include features described with reference to one or more figures, and may also include features described with reference to one or more other figures.

What is claimed is:

1. An integrated circuit device comprising:
 - a substrate including an active region defined by a device isolation layer, the substrate defining a gate trench extending across the active region;
 - a gate dielectric layer conformally covering an inner surface of the gate trench; and
 - a gate electrode filling the gate trench on the gate dielectric layer,
 wherein the gate electrode is composed of crystal grains of a single metal, and
 - a diagonal length of at least one of the crystal grains is greater than a horizontal width of a top surface of the active region that is in contact with the gate electrode.
2. The integrated circuit device of claim 1, wherein the single metal is tungsten (W), and
 - a diagonal length of each of the crystal grains is about 80 nm to about 100 nm.
3. The integrated circuit device of claim 2, wherein each of the crystal grains has a superlattice structure.
4. The integrated circuit device of claim 2, wherein the gate electrode that is in contact with the active region has seven or fewer grains of tungsten.
5. The integrated circuit device of claim 1, further comprising:
 - a gate barrier layer between the gate dielectric layer and the gate electrode, the gate barrier layer including a metal nitride.

6. The integrated circuit device of claim 1, wherein the gate dielectric layer and the gate electrode correspond to a word line buried in the substrate.
7. The integrated circuit device of claim 6, further comprising:
 - a capping insulation layer filling the gate trench above the gate dielectric layer and the gate electrode.
8. The integrated circuit device of claim 7, wherein the active region has a fin shape under the buried word line.
9. The integrated circuit device of claim 1, wherein the crystal grains comprise first crystal grains filling the gate trench, and second crystal grains between the first crystal grains and the gate dielectric layer, at least one of the second crystal grains smaller in grain size than at least one of the first crystal grains.
10. The integrated circuit device of claim 9, wherein the single metal is tungsten (W),
 - a diagonal length of each of the first crystal grains is about 80 nm to about 100 nm, and
 - the diagonal length of at least one of the first crystal grains is greater than the horizontal width of the top surface of the active region that is in contact with the gate electrode.
11. A method of manufacturing an integrated circuit device, the method comprising:
 - preparing a substrate including an active region having a fin shape defined by a device isolation layer;
 - forming a gate trench extending across the active region;
 - forming a gate dielectric layer conformally covering an internal surface of the gate trench; and
 - forming a gate electrode filling the gate trench on the gate dielectric layer,
 wherein the gate electrode is composed of crystal grains of a single metal, and
 - a diagonal length of each of the crystal grains is greater than a height of the active region that is in contact with the gate electrode.
12. The method of claim 11, wherein the single metal is tungsten (W), and the diagonal length of each of the crystal grains is about 80 nm to about 100 nm.
13. The method of claim 12, wherein the single metal is formed by a tungsten forming process using a source gas, the source gas including at least one of borane (BH₃), diborane (B₂H₆), monosilane (SiH₄), and tungsten hexafluoride (WF₆).
14. The method of claim 13, wherein the tungsten forming process comprises:
 - a nucleation layer forming process performed at a process temperature of about 250° C. or less; and
 - a bulk layer forming process performed at a process temperature of about 300° C. or less.
15. The method of claim 14, wherein each of the crystal grains has a superlattice structure, and the gate electrode that is in contact with the active region has seven or fewer grains of tungsten.
16. A method of manufacturing an integrated circuit device, the method comprising:
 - preparing a substrate including an active region having a fin shape defined by a device isolation layer;
 - forming a gate trench extending across the active region;
 - forming a gate dielectric layer conformally covering an internal surface of the gate trench; and

forming a gate electrode filling the gate trench on the gate dielectric layer,

wherein the gate electrode is composed of crystal grains of a single metal, and

a diagonal length of each of the crystal grains is greater than a horizontal width of a top surface of the active region that is in contact with the gate electrode.

17. The method of claim **16**, wherein the single metal is tungsten (W), and the diagonal length of each of the crystal grains is about 80 nm to about 100 nm.

18. The method of claim **17**, wherein the single metal is formed by a tungsten forming process using a source gas, the source gas including at least one of borane (BH_3), diborane (B_2H_6), monosilane (SiH_4), and tungsten hexafluoride (WF_6).

19. The method of claim **18**, wherein the tungsten forming process comprises:
a nucleation layer forming process performed at a process temperature of about 250°C . or less; and
a bulk layer forming process performed at a process temperature of about 300°C . or less.

20. The method of claim **19**, wherein each of the crystal grains has a superlattice structure, and the gate electrode that is in contact with the active region has seven or fewer grains of tungsten.

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